Jabatan Pembangunan Kemahiran
Kementerian Sumber Manusia, Malaysia

NATIONAL OCCUPATIONAL SKILLS STANDARD
(STANDARD KEMAHIRAN PEKERJAAN KEBANGSAAN)

G452-011-4:2025

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

DIAGNOSTIK ELEKTRIK AUTOMOTIF

LEVEL 4

Department of Skills Development (DSD)
Federal Government Administrative Centre
62530 PUTRAJAYA, MALAYSIA

NATIONAL OCCUPATIONAL SKILLS STANDARD

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

DIAGNOSTIK ELEKTRIK AUTOMOTIF

LEVEL 4

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Preface

Standard Definition

The National Occupational Skills Standard (NOSS) is a Standard document that outlines the **minimum** competencies required by a skilled worker working in Malaysia for a particular area and level of occupational, also the path to achieve the competencies. The competencies are based on the needs of employment, according to the career structure for the occupational area and developed by industry experts and skilled workers.

The National Competency Standard (NCS) is a Standard document that outlines the competencies required by a skilled worker in Malaysia.

Description of Standard Components

The document is divided into three (3) components which includes: -

Component I **Standard Practice**
This component is about the information related to occupational area including introduction to the industry, Standard requirements, occupational structure, levelling of competency, authority and industry requirements as a whole.

Component II **Standard Content**
This component is a reference to industry employers in assessing and improving the competencies that is required for a skilled worker. The competencies are specific to the occupational area. The component is divided into two (2) section which are the chart (Competency Profile Chart, CPC) and details of the competencies (Competency Profile, CP).

Component III **Curriculum of Competency Unit**
This component is a reference for the training personnel to identify training requirements, design the curriculum, and develop assessment. The training hours that included in this component is based on the recommendations by the Standard Development Committee (SDC). If there are modifications to the training hours, the Department provides the medium for discussion and consideration for the matter.

Abbreviation

1	4M	Man, Method, Machine and Material
2	ABS	Anti-lock Brake System
3	ADAS	Advance Driver Assistance System
4	ADTEC	*Advanced Technology Training Center*
5	BEV	Battery-powered Electric Vehicle
6	CP	Competency Profile
7	CPC	Competency Profile Chart
8	CPC_{PdP}	Competency Profile Chart for Teaching & Learning
9	CU	Competency Unit
10	CVT	Continuous Variable Transmission
11	DCT	Dual Clutch Transmission
12	D-Logger	Data-Logger
13	DOE	Department of Environment
14	DOSH	Department of Occupational Safety and Health
15	DSD	Department of Skills Development
16	DTC	Diagnostic Trouble Code
17	ECU	Electronic Control Unit
18	EEV	Energy Efficient Vehicle
19	EMS	Engine Management System
20	EPB	Electronic Parking Brake
21	EPS	Electrical Power Steering
22	ESP	Electronic Stability Programme
23	HVAC	Heating, Ventilation, and Air Conditioning
24	IKBTN	*Institut Kemahiran Tinggi Belia Negara*
25	ISO	International Organization for Standardization
26	JPS	Standard Development Committee
27	JTPS	Standard Technical Evaluation Committee
28	JTS	Standard Technical Committee

29	LiDAR	Light Detection And Ranging
30	MARii	Malaysia Automotive Robotics and IoT Institute
31	MIROS	Malaysian Institute of Road Safety Research
32	MOHR	Ministry of Human Resource
33	MOT	Ministry of Transport
34	MPKK	National Skills Development Council
35	MS	Malaysian Standard
36	MSC	Malaysian Skills Certificate
37	MSD	Malaysian Skills Diploma
38	MSIC	Malaysian Standard Industrial Classification
39	NCS	National Competency Standard
40	NDT	Non-Destructive Testing
41	NOSS	National Occupational Skills Standard
42	OAS	Occupational Area Structure
43	OS	Occupational Structure
44	OSHA	Occupational Safety and Health Act
45	PPE	Personnel Protective Equipment
46	Puspakom	*Pusat Pemeriksaan Kenderaan Berkomputer*
47	PWM	Pulse Width Modulation
48	SIRIM	Standard and Industrial Research Institute of Malaysia
49	SOP	Standard Operating Procedures
50	SRS	Supplemental Restraint System
51	SST	Special Service Tool
52	TPMS	Tyre Pressure Monitoring System
53	TVET	Technical Vocational Education and Training (TVET)
54	VPW	Variable Pulse Width

Glossary

1	Anti-lock Brake System (ABS)	Vehicle safety feature designed to prevent wheels from locking up during braking. ABS automatically modulates brake pressure to maintain wheel rotation, especially on slippery surfaces, enabling the driver to retain steering control.
2	Advance Driver Assistance System (ADAS)	Collection of electronic technologies in vehicles designed to aid drivers and enhance safety. ADAS includes features such as adaptive cruise control, lane departure warning, automatic emergency braking, and parking assistance, all of which help prevent accidents by alerting drivers to potential hazards or automatically intervening when necessary.
3	Battery-powered Electric Vehicle (BEV)	Type of electric vehicle that is fully powered by a rechargeable battery pack, with no internal combustion engine. BEVs rely solely on electricity stored in their batteries to operate, producing zero emissions during use. They are recharged via external electrical sources and are known for their quiet operation, energy efficiency, and environmental benefits.
4	Electronic Control Unit (ECU)	Critical onboard computer in vehicles that controls one or more electrical systems or subsystems, such as engine, transmission, and braking. The ECU processes data from various sensors and makes real-time adjustments to optimize vehicle performance and safety.
5	Energy Efficient Vehicle (EEV)	Vehicle designed with advanced technologies or alternative fuels to reduce fuel consumption and emissions. EEVs may include electric vehicles, hybrids, and other vehicles that meet high standards for energy efficiency and environmental impact.
6	Engine Management System (EMS)	System that controls and optimizes engine performance, efficiency, and emissions. The EMS monitors data from sensors and adjusts fuel injection, ignition timing, and other engine parameters to improve fuel economy and reduce emissions.
7	Electronic Parking Brake (EPB)	Electronically controlled brake system that replaces the traditional handbrake lever. Activated by a button, the EPB uses actuators to engage the parking brake, providing greater convenience and often integrating with other safety features.
8	Electric Power Steering (EPS)	Steering system that uses an electric motor to assist the driver, rather than a hydraulic pump. EPS improves fuel efficiency and allows for variable steering assistance, enhancing control at different speeds and reducing driver effort.
9	Electronic Stability Programme (ESP)	Safety system that improves vehicle stability by detecting and reducing skidding. ESP monitors wheel speed, steering angle, and other data to apply individual brakes or reduce engine power, helping prevent loss of control in slippery conditions.

10	Light Detection and Ranging (LiDAR)	Remote sensing technology that uses laser light to measure distances by calculating the time it takes for the light to return after bouncing off objects. LiDAR creates high-resolution, 3D maps of surroundings and is widely used in autonomous vehicles to detect obstacles and improve navigation and safety.
11	Supplemental Restraint System (SRS)	A safety system in vehicles designed to supplement seat belts in protecting passengers during collisions. The SRS includes airbags, seat belt pretensioners, and other components that are automatically deployed to reduce injury severity in the event of a crash.

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Acknowledgement

Director General of Department of Skills Development (DSD) would like to extend gratitude to the National Skills Development Council (MPKK), Standard Technical Committee (JTS), Standard Technical Evaluation Committee (JTPS), Standard Development Committee (JPS), and organisation and individuals who have been involved directly or indirectly for the contribution, persistence, and support in the development of this Standard until it is completed.

STANDARD PRACTICE

NATIONAL OCCUPATIONAL SKILLS STANDARD (NOSS) FOR:

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

1. Introduction

In the past, automotive electrical and electronic systems were simple, with basic wiring for lighting, ignition, and a limited number of gauges. Mechanical systems, rather than electronics, controlled most vehicle functions, and safety features were minimal. As vehicles evolved in the 1980s and 1990s, advancements like Electronic Control Units (ECUs) emerged, centralizing control over fuel injection, ignition timing, and other critical functions, significantly improving engine efficiency and emissions. The introduction of Anti-Lock Braking Systems (ABS) and airbags marked a pivotal shift toward enhanced safety and the integration of electronics in automotive systems.

Today, modern vehicles are equipped with an array of advanced electronics and mechatronics systems that significantly improve safety, efficiency, and convenience. Systems like Advanced Driver Assistance Systems (ADAS) now provide semi-autonomous capabilities, while Engine Management Systems (EMS) and electronic stability controls work seamlessly to optimize performance and safety. The future holds further integration of Artificial Intelligence (AI) and Vehicle-to-Everything (V2X) communication, paving the way for fully autonomous vehicles. Emerging technologies, such as solid-state batteries for electric vehicles and advanced powertrain mechatronics, aim to increase sustainability and reduce environmental impact. As the industry progresses, a stronger focus on smart mobility solutions, including intelligent infrastructure, personalized in-car experiences, and cloud-based software updates, making cars more connected, autonomous, and adaptable to user needs.

1.1 Occupation Overview

An Automotive Electrical Diagnostic is a specialized inspection and troubleshooting process aimed at identifying and resolving electrical issues within a vehicle. This service involves the use of advanced diagnostic tools and techniques to assess components like the battery, alternator, starter motor, wiring, fuses, sensors, and control modules. Modern vehicles are equipped with complex electrical systems that control functions ranging from lighting and power windows to infotainment, climate control, and safety systems. Electrical diagnose service professionals examine these systems to locate faults, which may include malfunctioning sensors, wiring issues, dead batteries, or failed modules. By conducting a thorough inspection and performing precise diagnostics, technicians can accurately identify the root causes of electrical problems, clear error codes, and restore proper function to all electrical components. This service ensures vehicle reliability, safety, and performance by maintaining a well-functioning electrical system and often includes preventive maintenance to minimize future issues.

A highly skilled automotive electrical master technician or expert is essential because modern vehicles rely heavily on complex electrical systems for everything from basic functions, like lighting and ignition, to advanced features, such as driver assistance systems (ADAS), infotainment, and engine control. As vehicles evolve, they integrate more sophisticated electronics, sensors, and computer systems that require specialized knowledge to diagnose, repair, and maintain. A master technician has the expertise to interpret diagnostic data accurately, troubleshoot intricate issues, and perform precise repairs that general technicians may not be equipped to handle. Their advanced skill set reduces the risk of misdiagnosis and ensures that repairs are done efficiently and correctly, minimizing vehicle downtime and repair costs. Additionally, a knowledgeable expert keeps up with the latest technology and industry standards, providing confidence to customers and supporting the overall safety, performance, and longevity of the vehicle.

1.2 Rationale of NOSS Development

The review of the existing National Occupational Skills Standard (NOSS), namely TP-310-4-Automotive Electrical Technologist, which was developed in 2010, is essential to ensure that the occupational standard remains relevant, effective, and aligned with industry standards and educational goals. Automotive Diagnostic (Electrical) is constantly evolving due to technological advancements, changing regulations, and emerging industry trends. A NOSS review helps ensure that the course content reflects the latest developments.

According to the skill category based on MASCO description, skilled workers consist of managers, professionals, technicians and associate professionals. The skilled and semi-skilled workers for all groups within G-452 Maintenance and Repair of Motor Vehicles are in high demand as the current manpower supply is in high shortage. Elementary/low-skilled workers are low in demand as there is a surplus of foreign workers to fill up the low-skilled workers segmentation in Jobs in Demand[1].

This NOSS outlines the competencies in Automotive Diagnostic (Electrical) as required by the industry and has been developed and documented following extensive collaboration across key Malaysian organizations. To meet the industry's requirements, the competencies outlined must follow a high standard and maintain consistency throughout the assessment process. This can only be done by stipulating a precise framework in which competencies must be assessed.

In this new NOSS all existing competency units were reviewed and reformed according to current industry practice. Industry panels decided that some competency units from the existing Malaysian Skills Diploma (MSC) shall be renamed and restructured in accordance with current and future practices in the industry. The matrix indicating the mapping of competencies between the new and revised NOSS is depicted in Table 1 below.

Duty 01 and 02 in TP-310-4 - Automotive Electrical Technologist have been reviewed and determined to be obsolete as they no longer align with the evolving requirements and priorities of the current industry landscape. These duties either do not reflect contemporary practices, have become redundant due to advancements in technology or methodologies, or are no longer relevant to the skill sets expected from professionals in this domain.

On the other hand, Duty 07, 08, and 09 have been shifted to Level 5 due to their complexity, criticality, and alignment with the higher-order competencies required at this level. These duties demand advanced skills, specialized knowledge, and

[1] Occupational Framework - Wholesale and Retail Trade and Repair of Motor Vehicles and Motorcycles - ISBN 978-967-0572-85-7

decision-making capabilities that are better suited for professionals operating at Level 5. This reassignment ensures that the competency framework reflects the industry's expectations and provides a clear progression path for individuals aspiring to excel in this field.

Table 1: Competency units mapping for TP-310-4 - Automotive Electrical Technologist to newly reviewed G452-011-4:2025 - Automotive Electrical Diagnostic

NOSS	G452-011-4:2025 - AUTOMOTIVE ELECTRICAL DIAGNOSTIC						NOTE
	COMPETENCY UNIT	G452-011-4:202X-C01 Perform automotive Engine Management System (EMS) diagnosis	G452-011-4:202X-C02 Perform automotive body electrical system diagnosis	G452-011-4:202X-C03 Perform automotive chassis electrical diagnosis	G452-011-4:202X-C04 Perform automotive powertrain system diagnosis	*G452-011-4:202X-C05 Perform automotive Advance Driver Assistance System (ADAS) diagnosis	
TP-310-4 - AUTOMOTIVE ELECTRICAL TECHNOLOGIST	Duty 01 - Perform Automotive Electrical System Design						Duty 01 & 02 are obsolete because they are irrelevant to current industry requirements.
	Duty 02 - Perform Automotive Electrical System Parts & Fixtures Development Activities						
	Duty 03 - Perform Automotive Electrical & System Diagnosis	X	X	X	X		-
	Duty 04 - Perform Automotive Electrical Auxiliary & Safety Comfort System Diagnosis		X	X			-
	Duty 05 - Perform Automotive Electrical Engine Management & Control System Diagnosis	X					-
	Duty 06 - Perform Automative Electrical System Performance Evaluation Activities	X	X	X	X		-
	Duty 07 - Organize Automotive Electrical Maintenance Activities						Duty 07, 08 & 09 are covered at Level 5 due to the nature of competency related to industry requirements.
	Duty 08 - Perform Automotive Electrical System Quality Control Activities						
	Duty 09 - Perform Managerial Functions						

X = The previous competency unit was retained/covered in this NOSS.
*New competency unit for this NOSS.

1.3 Rationale of Occupational Structure and Occupational Area Structure

Automotive Diagnostic (Electrical) is classified in (G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles and (452) Maintenance and Repair of Motor Vehicles in accordance with Malaysia Standard Industry Classification (MSIC 2008). This classification also matches the Malaysia Standard Classification of Occupations (MASCO), a national benchmark for classifying occupations in the country's employment structure. MASCO has been developed following the International Standard Classification of Occupations (ISCO) with changes and modifications to meet the needs of the country[2].

As a result of the Occupational Framework (G45) - Wholesale and Retail Trade and Repair of Motor Vehicles and Motorcycles study conducted in 2018, together with expert panel members from various organizations, identified a total of 31 sub-sectors and 104 job titles. Based on the survey findings, the survey respondents highlighted the skills in demand for G452 - Maintenance and Repair of Motor Vehicles, including diagnostic skills and troubleshooting/problem-solving skills. The areas developed include Repair/ Maintenance (Light / Heavy), Energy Efficiency Vehicle – Hybrid and Technical Support. Under the Technical Support area, job titles - Head of After Sales, Workshop Manager and Technical Support Executive are suggested for Level 6, 5 and 4, respectively[3].

Considering these findings, NOSS development panels came with the Occupational Structure and Occupational Area Structure for Maintenance and Repair of Motor Vehicles, which covers Electrical System, Mechanical System and Operation as illustrated in Figure 1 and Figure 2, respectively. The scope of this NOSS is limited to the diagnosis of the electrical system only and does not include the mechanical system.

Based on the Occupational Area Structure, the person who is competent in Level 3 can choose either the After Sales - Service Operation Level 4 or Automotive Diagnostic (Electrical) Level 4.

1.4 Regulatory/Statutory Body Requirements Related to Occupation

The occupation of Automotive Diagnostic (Electrical) Level 4 in Malaysia generally subjected to the following act and regulatory requirements governed by:

a) Ministry of Human Resource:
   - i) Department of Occupational Safety and Health (DOSH).
     - Occupational Safety and Health 1994 Act (Act 514).
   - ii) Department of Labour Peninsular Malaysia:
     - Employment Act 1955 (Act 265).
   - iii) Department of Labour Sabah:
     - Sabah Labour Ordinance 1949 (Sabah Cap. 67).
   - iv) Department of Labour Sarawak:

[2] https://emasco.mohr.gov.my/masco/2149-16
[3] Occupational Framework - Wholesale and Retail Trade and Repair of Motor Vehicles and Motorcycles - ISBN 978-967-0572-85-7

- Sarawak Labour Ordinance 1952 (Sarawak Cap. 76).

b) Ministry of Natural Resources and Environmental Sustainability:
   i) Department of Environment (DOE)
      - Environmental Quality Act 1974 (Act 127).

c) Ministry of Domestic Trade and Cost of Living:
   i) Consumer Protection Act 1999 (Act 599).
   ii) Trade Descriptions Act 2011 (Act No. 730).

d) Department of Standard Malaysia:
   i) MS 2696:2018 Motor vehicle aftermarket - Service and spare parts (2S) - Code of practice.

1.5 Occupational Prerequisite

The occupational prerequisites for this job position are as follows:

a) Diploma in Automotive / Autotronic / Electrical / Electronic / Mechanical engineering or technology; and
b) At least 5 years of relevant experience determined by company policies and requirements.

1.6 General Training Prerequisite for Malaysian Skills Certification System

The minimum requirements set forth before enrolling for this course is that the candidates must possess a Malaysian Skills Certificate (MSC) Level 3 in either of the following fields:

a) G452-002-3:2018 Light Vehicle – Diagnose Service; or
b) G452-007-3:2019 Electric and Hybrid Car Servicing; or
c) G452-001-3:2017 Commercial Vehicle – Troubleshooting Service; or
d) G452-008-3:2020 Motorsports Vehicle (Four-Wheel) Preparation and Maintenance; or
e) G452-010-3:2023 Battery Electric Vehicle (BEV) Diagnostic and Rectification; or
f) TP-331-3:2014 Electric Bus Operation; or
g) TP-302-3 Motorsport Technician (Electrical & Electronics); or
h) TP-100-3:2013 Earth Moving Equipment Maintenance; or
i) AT01: *Automotif Mekatronik*.

2. Occupational Structure (OS)

Section	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
Group	(452) Maintenance and Repair of Motor Vehicles		
Area	Maintenance and Repair of Motor Vehicles		
	Electrical System	Mechanical System	Operation
Level 5	Technical Executive	Technical Executive	Service / Operation Manager
Level 4	Master Technician / Foreman	Master Technician / Foreman	Service / Operation Executive
Level 3	Passenger Vehicle Maintenance & Service Diagnostic Technician		
Level 2	Motor Vehicle Maintenance & Service Senior Technician		
Level 1	Motor Vehicle Repair and Maintenance Technician		

Figure 1: Occupational Structure for Maintenance and Repair of Motor Vehicles

3. Occupational Area Structure (OAS)

Section	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
Group	(452) Maintenance and Repair of Motor Vehicles		
Area	Maintenance and Repair of Motor Vehicles		
	Electrical System	Mechanical System	After Sales Service
Level 5	Automotive Electrical Diagnostic	Automotive Mechanical Diagnostic Service	Motor Vehicles - After Sales Service Management
Level 4	Automotive Electrical Diagnostic	Automotive Mechanical Diagnostic	Motor Vehicles - After Sales Service Operation
Level 3	Light Vehicle – Diagnose Service		
Level 2	Light Vehicle – Repair Service		
Level 1	Embedded to L2		

Figure 2: Occupational Area Structure for Maintenance and Repair of Motor Vehicles

4. Definition of Competency Levels

The NOSS is developed for various occupational areas. Below is a guideline of each NOSS Level as defined by the Department of Skills Development, Ministry of Human Resources, Malaysia.

Level 1: Competent in performing a range of varied work activities, most of which are routine and predictable.

Level 2: Competent in performing a significant range of varied work activities, performed in a variety of contexts. Some of the activities are non-routine and required individual responsibility and autonomy.

Level 3: Competent in performing a broad range of varied work activities, performed in a variety of contexts, most of which are complex and non-routine. There is considerable responsibility and autonomy and control or guidance of others is often required.

Level 4: Competent in performing a broad range of complex technical or professional work activities performed in a wide variety of contexts and with a substantial degree of personal responsibility and autonomy. Responsibility for the work of others and allocation of resources is often present.

Level 5: Competent in applying a significant range of fundamental principles and complex techniques across a wide and often unpredictable variety of contexts. Very substantial personal autonomy and often significant responsibility for the work of others and for the allocation of substantial resources features strongly, as do personal accountabilities for analysis, diagnosis, planning, execution and evaluation.

5. Award of Certificate

The Director General may award, to any person upon conforming to the Standards the following skills qualifications as stipulated under the National Skills Development Act 2006 (Act 652):

a) Malaysian Skills Diploma (MSD); or
b) Statements of Achievement.

6. Occupational Competencies

The Automotive Electrical Diagnostic Level 4 personnel is competent in performing the following core competencies:

a) Perform automotive Engine Management System (EMS) diagnosis;
b) Perform automotive body electrical system diagnosis;
c) Perform automotive chassis electrical diagnosis;
d) Perform automotive powertrain system diagnosis; and
e) Perform automotive Advance Driver Assistance System (ADAS) diagnosis.

7. Work Conditions

An automotive electrical master technician typically works in a dynamic and hands-on environment, such as a repair shop, dealership, or specialized automotive service centre. Work conditions may include standing for extended periods, bending, or working in tight spaces to access various vehicle components. Given the nature of the job, technicians regularly handle specialized diagnostic tools, electrical testing equipment, and sometimes heavy parts or components.

This NOSS involves low voltage for light vehicles typically 12V or 24V systems such as interior lighting, power windows, infotainment system, car alarm, USB for charging and does not include high voltage systems that are regulated under the Energy Commission Act. However, work with low-voltage systems requiring a thorough understanding of safety protocols to prevent electrical hazards. The job often demands precision, focus, and problem-solving skills to troubleshoot complex systems accurately. Additionally, because automotive technology is continually evolving, the role requires ongoing training and education, so technicians often work in environments that encourage skills development and stay updated with industry advancements. The work can be fast-paced and occasionally high-pressure, especially when dealing with complex or urgent repairs. Still, it is rewarding for those with a strong interest in automotive technology and problem-solving.

8. Employment Prospects

The employment prospects for automotive electrical technicians or experts are strong and expected to grow due to the increasing complexity of modern vehicles, which rely on advanced electrical and electronic systems. With the rise of electric vehicles (EVs), hybrid technology, and sophisticated driver assistance systems, there is a high demand for technicians skilled in diagnosing, repairing, and maintaining these advanced systems.

Malaysia's total automotive industry volume is expected to reach 1.22 million units in 2030 from 600,000 units in 2020[4]. This increase is driven by, among others, the growth of the after-sales market, particularly the expansion of the market for spare parts, repairs and maintenance services. During the period of the NIMP 2030, in line with NAP 2020, the industry will focus on ensuring the development of human capital keeps pace with advancements in current and future automotive technology[5].

According to development panels, the average monthly starting salary for an Automotive Master Technician in Malaysia ranges from RM 2,000 to RM 2,500. On the other hand, the ILMIA report shows that under Motor Vehicle Technicians (MASCO: 3154), the Motor Vehicle Technicians (service and repair vehicle engines) earn RM 4,883 (Mean Salary) or RM 3,250 (Median Salary)[6].

The actual salary offered depends on the industrial sector, the size of the employer's operations, skill level, and work experience. In addition, employers provide other incentives (allowances).

Automotive manufacturers, dealerships, repair shops, and specialized EV service centres continuously seek qualified experts who understand traditional vehicle electrical systems and newer technologies. As the automotive industry continues to shift toward electric and autonomous vehicles, technicians with expertise in EV systems, battery management, high-voltage components, and electronic diagnostics will be especially sought. This demand for specialized skills presents promising opportunities for career advancement, higher wages, and job stability for automotive electrical technicians who invest in continuous learning and certification.

9. Up Skilling Opportunities

Pursuing these upskilling opportunities not only strengthens technical knowledge but also prepares automotive electrical technicians for future advancements in the industry, positioning them as experts in their field. Here are some key upskilling opportunities for automotive electrical technicians or experts:

a) Advanced Driver Assistance Systems (ADAS) Training: As more vehicles integrate ADAS features, training in ADAS calibration, sensor alignment, and diagnostic techniques is essential. Programs focused on ADAS systems provide the knowledge to service and calibrate technologies like lane-keeping assist, adaptive cruise control, and collision avoidance.

b) Manufacturer-Specific Training: Many automotive brands offer specialized training for their vehicle systems, covering proprietary technology and diagnostic techniques. Certifications from automotive brands provide deeper knowledge of brand-specific systems and tools, making technicians more versatile and marketable.

c) Advanced Electrical and Electronics Courses: Courses in advanced automotive electronics, such as microcontroller diagnostics, circuit board repairs, and sensor

[4] National Automotive Policy 2020 (NAP 2020)
[5] Malaysia New Industrial Master Plan 2030 (NMP 2030) - Automotive Industry (Ministry Of Investment, Trade and Industry)
[6] Q4 2023, Job Market Insights, ILMIA https://www.ilmia.gov.my/index.php/en/bda-jmi -

technologies, help technicians improve their understanding of vehicle electronics and stay ahead of evolving technology.

d) Diagnostic Software Proficiency: Gaining expertise in popular diagnostic software, , or OEM-specific diagnostic tools, allows technicians to accurately interpret error codes, run system tests, and streamline repairs. Proficiency with these tools is increasingly essential in today's diagnostic-heavy work.

e) High-Voltage Safety and Certification: For technicians working with electric and hybrid vehicles, training in high-voltage safety protocols and handling is essential to prevent accidents and ensure compliance with safety standards.

f) Electric and Hybrid Vehicle Certification: With the rise of electric and hybrid vehicles, certifications such as the ASE L3 (Light Duty Hybrid/Electric Vehicle Specialist) or manufacturer-specific EV training can help technicians gain expertise in high-voltage systems, battery diagnostics, and EV-specific repairs.

g) Soft Skills and Customer Communication: Courses in communication and customer service help technicians explain complex issues to clients, enhancing trust and improving overall customer satisfaction.

10. Organisation Reference for Sources of Additional Information

The following organisations can be referred as sources of additional information which can assist in defining the document's contents.

a) Ministry of Domestic Trade and Cost of Living,
No. 13, Persiaran Perdana,
Presint 2, 62623 Putrajaya, MALAYSIA.
Tel: 60380008000
Fax: 60388825762
Web: www.kpdn.gov.my
Email: e-aduan.kpdnkk.gov.my

b) Malaysia Automotive Robotics and IoT Institute (MARii),
Ministry of International Trade and Industry (MITI)
Block 2280, Jalan Usahawan 2, Cyber 6
63000 Cyberjaya, Selangor, MALAYSIA.
Tel: 60383187742
Fax: 60383187743
Web: www.marii.my
Email: autoaftermarket@mai.org.my

c) Malaysian Institute of Road Safety Research (MIROS)
Lot 125-135, Jalan TKS 1, Taman Kajang Sentral,
43000 Kajang, Selangor, MALAYSIA.
Tel: 603 8924 9200
Fax: 603 8733 2005
Web: www.miros.gov.my
Email: inquiry@miros.gov.my

d) Road Transport Department,
Ministry of Transport Malaysia (MOT)
Aras 3-5, No. 26, Jalan Tun Hussien, Presint 4,
Pusat Pentadbiran Kerajaan Persekutuan,
62100 WP Putrajaya, MALAYSIA.
Tel: 60380008000
Web: www.jpj.gov.my
Email: aduantrafik@jpj.gov.my

e) Department of Environment (DOE),
Ministry of Energy, Science, Technology,
Environment & Climate Change (MESTECC)
Block 4G3, Aras1-4, Podium 2, 3, 25,
Persiaran Perdana,
62574 Putrajaya, MALAYSIA.
Tel: 60388712000
Web: www.nre.gov.my
Email: aduan_k@doe.gov.my

f) PUSPAKOM Sdn. Bhd. (285985-U),
Level 3 & 4, Wisma DRB-HICOM,
No 2, Jalan Usahawan U1/8, Seksyen U1,
40150 Shah Alam, Selangor, MALAYSIA.
Tel: 60320527474
Web: www.puspakom.com.my
Email: customerservice@puspakom.com.my

11. Standard Technical Evaluation Committee

NO	NAME	POSITION & ORGANISATION
CHAIRMAN		
1	Nor Asmah binti Daud	Principal Assistant Director Department of Skills Development
EVALUATION PANEL		
1	Mohamad Sufian bin Ngah Mat Noh	Automotive Technology Instructor lnstitut Latihan Perindustrian (ILP) Kuala Lumpur
2	Dato Shamsuddin bin Amran	Group Head After Sales Bermaz Auto Berhad
3	Chung Fook Chen	Senior Manager II Mazda Technical Training
4	Ben Lee Yew Cheong	Chairman EJK Car Solution & Lubricant (M) Sdn. Bhd.
SECRETARIAT		
1	Mohd Shahrol @ Shukor bin Salleh	Senior Assistant Director Department of Skills Development

12. Standard Development Committee

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

NO	NAME	POSITION & ORGANISATION
DEVELOPMENT PANEL		
1	Hung Yew Kim	Section Manager Proton Global Services Sdn. Bhd.
2	Leong Khing Sang	Service Manager CSH Automobiles Sdn. Bhd.
3	Mohd Nor Akmal bin Anuar	Technical Training Manager Bermaz Auto Berhad
4	Muhammad Nur Salam bin Zainuddin	Assistant Manager Bermaz Auto Berhad
5	Tc. Zabidi bin Zamzamin	Director Iman Education and Technology Sdn. Bhd.
6	Tc. Mohamad Zaki bin Abdul Hamed	Senior Manager Technical and Training EZ Ventures International Sdn. Bhd.
7	Faszlin binti Rahim	Assistant Manager Tan Chong Education Services Sdn. Bhd.
8	Zaidi bin Pandak	Technical Trainer KWT Technology Sdn. Bhd.
9	Tc. Tan Kim Heng	Automotive Technical Trainer AD Tech Consultant & Resources
10	Tc. Mohammad Fairil bin Din @ Samsudin	Training Executive HICOM Automotive Manufacturers Malaysia Sdn. Bhd.
11	Nur Muhammad bin Haji Mohd Rajoli	Head of Section Automotive Technology IKTBN Dusun Tua
12	Ts. Ting Swee Keong	Vocational Training Officer ADTEC Shah Alam
13	Mohd Khaleff bin Md Shaharill	Vocational Training Officer IKTBN Dusun Tua
FACILITATOR		
1	Ts. Mohd Razali Bin Md Yunos	CIAST/PPL/FDS-0323/2019 IMZA Oracle Sdn. Bhd.

STANDARD CONTENT

NATIONAL OCCUPATIONAL SKILLS STANDARD (NOSS) FOR:

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

13. Competency Profile Chart (CPC)

SECTION	(G) WHOLESALE AND RETAIL TRADE; REPAIR OF MOTOR VEHICLES AND MOTORCYCLES		
GROUP	(452) MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
AREA	MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
NOSS TITLE	AUTOMOTIVE ELECTRICAL DIAGNOSTIC		
NOSS LEVEL	FOUR (4)	NOSS CODE	G452-011-4:2025

↤COMPETENCY UNIT↦		↤WORK ACTIVITIES↦			
CORE	PERFORM AUTOMOTIVE ENGINE MANAGEMENT SYSTEM (EMS) DIAGNOSIS G452-011-4:2025-C01	DIAGNOSE IGNITION SYSTEM G452-011-4:2025-C01-W01	DIAGNOSE STARTING SYSTEM G452-011-4:2025-C01-W02	DIAGNOSE CHARGING SYSTEM G452-011-4:2025-C01-W03	DIAGNOSE ELECTRIC FUEL DELIVERY SYSTEM G452-011-4:2025-C01-W04
		DIAGNOSE ENGINE CONTROL SYSTEM G452-011-4:2025-C01-W05			

	↤COMPETENCY UNIT↦	↤WORK ACTIVITIES↦			
CORE	PERFORM AUTOMOTIVE BODY ELECTRICAL SYSTEM DIAGNOSIS G452-011-4:2025-C02	DIAGNOSE LIGHTING AND INSTRUMENT PANEL SYSTEM G452-011-4:2025-C02-W01	DIAGNOSE WASHER AND WIPER SYSTEM G452-011-4:2025-C02-W02	DIAGNOSE POWER WINDOW AND MIRROR SYSTEM G452-011-4:2025-C02-W03	DIAGNOSE INFOTAINMENT SYSTEM G452-011-4:2025-C02-W04
		DIAGNOSE SECURITY AND LOCK SYSTEM G452-011-4:2025-C02-W05	DIAGNOSE SUPPLEMENTAL RESTRAINT SYSTEM (SRS) G452-011-4:2025-C02-W06	DIAGNOSE ELECTRICAL HEATING, VENTILATION AND AIR CONDITIONING SYSTEM (HVAC) G452-011-4:2025-C02-W07	
	PERFORM AUTOMOTIVE CHASSIS ELECTRICAL DIAGNOSIS G452-011-4:2025-C03	DIAGNOSE ELECTRIC POWER STEERING (EPS) SYSTEM G452-011-4:2025-C03-W01	DIAGNOSE ANTI-LOCK BRAKING SYSTEM (ABS) G452-011-4:2025-C03-W02	DIAGNOSE ELECTRONIC PARKING BRAKE (EPB) G452-011-4:2025-C03-W03	DIAGNOSE ELECTRONIC STABILITY PROGRAMME (ESP) SYSTEM G452-011-4:2025-C03-W04

	↤COMPETENCY UNIT↦	↤WORK ACTIVITIES↦			
CORE		DIAGNOSE TYRE PRESSURE MONITORING SYSTEM (TPMS) G452-011-4:2025-C03-W05			
CORE	PERFORM AUTOMOTIVE POWERTRAIN SYSTEM DIAGNOSIS G452-011-4:2025-C04	DIAGNOSE AUTOMATIC TRANSMISSION / TRANSAXLE ELECTRICAL SYSTEM G452-011-4:2025-C04-W01	DIAGNOSE CONTINUOUS VARIABLE TRANSMISSION (CVT) ELECTRICAL SYSTEM G452-011-4:2025-C04-W02	DIAGNOSE DUAL CLUTCH TRANSMISSION (DCT) ELECTRICAL SYSTEM G452-011-4:2025-C04-W03	
CORE	PERFORM AUTOMOTIVE ADVANCE DRIVER ASSISTANCE SYSTEM (ADAS) DIAGNOSIS G452-011-4:2025-C05	DIAGNOSE RADAR SYSTEM G452-011-4:2025-C05-W01	DIAGNOSE CAMERA SYSTEM G452-011-4:2025-C05-W02	DIAGNOSE ULTRASONIC SENSOR SYSTEM G452-011-4:2025-C05-W03	DIAGNOSE LIGHT DETECTION AND RANGING (LIDAR) SYSTEM G452-011-4:2025-C05-W04

14. Competency Profile (CP)

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
NOSS LEVEL	Four (4)	NOSS CODE	G452-011-4:2025

CU TITLE & CU CODE	Perform automotive Engine Management System (EMS) diagnosis. G452-011-4:2025-C01
CU DESCRIPTOR	Perform automotive Engine Management System (EMS) diagnosis to accurately troubleshoot, identify, and resolve issues within the engine's electronic control systems. This includes understanding how EMS components—like fuel injectors, ignition coils, sensors, and the ECM (Engine Control Module) - work together to regulate engine performance. A competent diagnostician can interpret diagnostic trouble codes (DTCs), analyse sensor data, and use specialized tools like OBD-II scanners and multimeters to pinpoint the source of issues. The person who is competent in this CU should be able to diagnose ignition system, diagnose starting system, diagnose charging system, diagnose electric fuel delivery system and diagnose engine control system. The outcome of this CU is issues within the EMS accurately diagnosed and resolved, ensuring the vehicle runs safely, efficiently and reliably with optimal power, fuel economy, and emissions compliance.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
1. Diagnose ignition system.	1.1 Review technical investigation and service information report. 1.2 Locate ignition system fault. 1.3 Troubleshoot ignition system fault. 1.4 Rectify ignition system fault. 1.5 Test ignition system.	1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (no start, misfire, rough idling, wiring issue) inspected and electrical components of ignition system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, ignition system wiring diagram interpreted, malfunction symptom of ignition system detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	1.6 Produce ignition system diagnostic report.	1.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 1.5 Ignition system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
2. Diagnose starting system.	2.1 Review technical investigation and service information report. 2.2 Locate starting system fault. 2.3 Troubleshoot starting system fault. 2.4 Rectify starting system fault. 2.5 Test starting system. 2.6 Produce starting system diagnostic report.	2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (cranking issues, wiring issue) inspected and electrical components of starting system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, starting system wiring diagram interpreted, malfunction symptom of starting system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 2.5 Starting system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
3. Diagnose charging system.	3.1 Review technical investigation and service information report. 3.2 Locate charging system fault. 3.3 Troubleshoot charging system fault. 3.4 Rectify charging system fault. 3.5 Test charging system. 3.6 Produce charging system diagnostic report.	3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (charging issue – low charge, over charge) inspected and electrical components of charging system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, charging system wiring diagram interpreted, malfunction symptom of charging system detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		3.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 3.5 Charging system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin, warranty report prepared according to company policy.
4. Diagnose electric fuel delivery system.	4.1 Review technical investigation and service information report. 4.2 Locate electric fuel delivery system fault. 4.3 Troubleshoot electric fuel delivery system fault. 4.4 Rectify electric fuel delivery system fault. 4.5 Test electric fuel delivery system. 4.6 Produce electric fuel delivery system diagnostic report.	4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (cranking (starting issue, electric fuel pump issue, fuel injector issue) inspected and electrical components of ignition system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, electric fuel delivery system wiring diagram interpreted, malfunction symptom of electric fuel delivery system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to wiring diagram and manufacturer standard. 4.5 Electric fuel delivery system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
5. Diagnose engine control system.	5.1 Review technical investigation and service information report. 5.2 Locate engine control system fault. 5.3 Troubleshoot engine control system fault.	5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (malfunction indicator lamp (MIL), starting issue, misfire issue, emission issue) inspected and electrical components of engine control system tested according to manufacturer diagnostic procedure. 5.3 Error code identified, engine control system wiring diagram, signal protocol (Pulse Width Modulation (PWM), Variable Pulse Width VPW,

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	5.4 Rectify engine control system fault. 5.5 Test engine control system. 5.6 Produce engine control system diagnostic report.	KWP, K-line, L-line) interpreted, malfunction symptom of engine control system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 5.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to wiring diagram and manufacturer standard. 5.5 Engine control system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

CU TITLE & CU CODE	Perform automotive body electrical system diagnosis. G452-011-4:2025-C02
CU DESCRIPTOR	Perform automotive body electrical system diagnosis describes to accurately troubleshoot and repair electrical issues within body systems such as interior and exterior lighting, climate control, infotainment, power seats, and keyless entry. This requires a solid understanding of body electrical systems, proficiency in reading wiring diagrams, and the ability to use diagnostic tools like multimeters and OBD scanners to pinpoint and resolve faults. A competent personnel can systematically identify the root cause of issues—whether from faulty wiring, blown fuses, or malfunctioning control modules—ensuring that all body electrical systems function reliably, enhancing comfort, safety, and convenience for the vehicle's occupants. The person who is competent in this CU should be able to diagnose lighting and instrument panel system, diagnose washer and wiper system, diagnose power window and mirror system, diagnose infotainment system, diagnose security and lock system, diagnose SRS and diagnose electrical HVAC system. The outcome of this CU is fully operational and reliable body systems, including lighting, climate control, infotainment, power windows, and keyless entry, enhancing both driver and passenger comfort and convenience. By accurately identifying and addressing issues—such as wiring faults, blown fuses, or malfunctioning control modules—the diagnosis restores full functionality to these systems, eliminates any related warning lights, and ensures smooth operation across all electrical features. This level of precision minimizes future electrical issues, reduces repair costs, and contributes to an overall improved driving experience, ensuring the vehicle's body electrical components work seamlessly.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
1. Diagnose lighting and instrument panel system.	1.1 Review technical investigation and service information report. 1.2 Locate lighting and instrument panel system fault. 1.3 Troubleshoot lighting and instrument panel system fault. 1.4 Rectify lighting and instrument panel system fault.	1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (wiring issue, lighting issue, indicator issue, switch issue, body control module issue) inspected and electrical components of lighting and instrument panel system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, lighting and instrument panel system wiring diagram interpreted, malfunction symptom of lighting and instrument

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	1.5 Test lighting and instrument panel system. 1.6 Produce lighting and instrument panel system diagnostic report.	panel system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component repair work / service / replacement and programming carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 1.5 Lighting and instrument panel system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
2. Diagnose washer and wiper system.	2.1 Review technical investigation and service information report. 2.2 Locate washer and wiper system fault. 2.3 Troubleshoot washer and wiper system fault. 2.4 Rectify washer and wiper system fault. 2.5 Test washer and wiper system. 2.6 Produce washer and wiper system diagnostic report.	2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (wiring issue, motor issue, body control module issue) inspected and electrical components of washer and wiper system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, washer and wiper system wiring diagram interpreted, malfunction symptom of washer and wiper system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 2.5 Washer and wiper system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
3. Diagnose power window and mirror system.	3.1 Review technical investigation and service information report. 3.2 Locate power window and mirror system fault. 3.3 Troubleshoot power window and mirror system fault.	3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (wiring issue, motor issue, body control module issue, rear demister) inspected and electrical components of power window and mirror system tested according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	3.4 Rectify power window and mirror system fault. 3.5 Test power window and mirror system. 3.6 Produce power window and mirror system diagnostic report.	3.3 Error code identified, power window and mirror system wiring diagram interpreted, malfunction symptom of power window and mirror system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement and programming carried out; fault code erased according to manufacturer diagnostic procedure. 3.5 Power window and mirror system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
4. Diagnose infotainment system.	4.1 Review technical investigation and service information report. 4.2 Locate infotainment system fault. 4.3 Troubleshoot infotainment system fault. 4.4 Rectify infotainment system fault. 4.5 Test infotainment system. 4.6 Produce infotainment system diagnostic report.	4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, software issue, signal issue, display issue, connectivity issue) inspected and electrical components of infotainment system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, infotainment system wiring diagram interpreted, malfunction symptom of infotainment system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component service work / replacement and programming carried out; fault code erased according to manufacturer diagnostic procedure. 4.5 Infotainment system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
5. Diagnose security and lock system.	5.1 Review technical investigation and service information report. 5.2 Locate security and lock system fault. 5.3 Troubleshoot security and lock system fault.	5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (starting issue, communication issue, wiring issue, alarm issue) inspected and electrical components of security and lock system tested according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	5.4 Rectify security and lock system fault. 5.5 Test security and lock system. 5.6 Produce security and lock system diagnostic report.	5.3 Error code identified, security and lock system wiring diagram interpreted, malfunction symptom of security and lock system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 5.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 5.5 Security and lock system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
6. Diagnose Supplemental Restraint System (SRS).	6.1 Review technical investigation and service information report. 6.2 Locate supplemental restraint system fault. 6.3 Troubleshoot supplemental restraint system fault. 6.4 Rectify supplemental restraint system fault. 6.5 Test supplemental restraint system. 6.6 Produce supplemental restraint system diagnostic report.	6.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 6.2 Symptoms (wiring issue, warning light, sensor issue) inspected and electrical components of supplemental restraint system tested according to manufacturer diagnostic procedure. 6.3 Error code identified, supplemental restraint system wiring diagram interpreted, malfunction symptom of supplemental restraint system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 6.4 Component replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 6.5 supplemental restraint system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 6.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
7. Diagnose Electrical Heating, Ventilation and Air Conditioning system (HVAC).	7.1 Review technical investigation and service information report. 7.2 Locate HVAC system fault. 7.3 Troubleshoot ignition system fault. 7.4 Rectify HVAC system fault. 7.5 Test HVAC system. 7.6 Produce HVAC system diagnostic report.	7.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 7.2 Symptoms (wiring issue, sensor issue, motor issue, compressor issue, climate control panel issue) inspected and electrical components of HVAC system tested according to manufacturer diagnostic procedure. 7.3 Error code identified, HVAC system wiring diagram interpreted, malfunction symptom of HVAC system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 7.4 Component repair work / service / replacement and programming carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 7.5 HVAC system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 7.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

CU TITLE & CU CODE	Perform automotive chassis electrical diagnosis. G452-011-4:2025-C03
CU DESCRIPTOR	Perform automotive chassis electrical diagnosis to effectively troubleshoot and resolve electrical issues within the vehicle's chassis systems. It involves understanding electrical principles, interpreting wiring diagrams, and using diagnostic tools like multimeters and scan tools to pinpoint faults accurately. A competent personnel can systematically identify issues—whether due to faulty wiring, connections, fuses, or sensors—and apply precise solutions, ensuring the vehicle's electrical systems operate reliably and safely while enhancing overall performance and driver satisfaction. The person who is competent in this CU should be able to diagnose Electric Power Steering (EPS), diagnose Anti-lock braking System (ABS), diagnose Electronic Parking Brake (EPB), diagnose Electronic Stability Programme (ESP and diagnose Tyre Pressure Monitoring System (TPMS). The outcome of this CU is all electrical systems within the chassis function reliably and efficiently by restoring full electrical functionality, eliminates any warning lights, and improves safety and convenience features for the driver. This helps prevent recurring electrical problems, reduces unnecessary battery drain, and contributes to a stable and dependable electrical system, enhancing the overall vehicle performance and driver confidence.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
1. Diagnose Electric Power Steering (EPS) system.	1.1 Review technical investigation and service information report. 1.2 Locate electric power steering system fault. 1.3 Troubleshoot electric power steering system fault. 1.4 Rectify electric power steering system fault. 1.5 Test electric power steering system. 1.6 Produce electric power steering system diagnostic report.	1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (steering issue, starting issue, wiring issue, indicator, sensor) inspected and electrical components of electric power steering system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, electric power steering system wiring diagram interpreted, malfunction symptom of electric power steering system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		1.5 Electric power steering system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
2. Diagnose Anti-lock Braking System (ABS)	2.1 Review technical investigation and service information report. 2.2 Locate anti-lock brake system fault. 2.3 Troubleshoot anti-lock brake system fault. 2.4 Rectify anti-lock brake system fault. 2.5 Test anti-lock brake system. 2.6 Produce anti-lock brake system diagnostic report.	2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (wiring issue, indicator, brake issue, sensor issue) inspected and electrical components of anti-lock brake system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, Anti-lock brake system wiring diagram interpreted, malfunction symptom of Anti-lock brake system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 2.5 Anti-lock brake system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
3. Diagnose Electronic Parking Brake (EPB).	3.1 Review technical investigation and service information report. 3.2 Locate EPB system fault. 3.3 Troubleshoot ignition system fault. 3.4 Rectify EPB system fault. 3.5 Test EPB system. 3.6 Produce EPB system diagnostic report.	3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (wiring issue, indicator, brake issue, sensor issue, motor issue, switch issue) inspected and electrical components of EPB system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, EPB system wiring diagram interpreted, malfunction symptom of EPB detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement, programming and calibration carried out; fault code erased according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		3.5 EPB system functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
4. Diagnose Electronic Stability Programme (ESP) system.	4.1 Review technical investigation and service information report. 4.2 Locate electronic stability programme system fault. 4.3 Troubleshoot ignition system fault. 4.4 Rectify electronic stability programme system fault. 4.5 Test electronic stability programme system. 4.6 Produce electronic stability programme system diagnostic report.	4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, indicator, brake issue, steering angle sensor issue, sensor issue) inspected and electrical components of electronic stability programme system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, electronic stability programme system wiring diagram interpreted, malfunction symptom of electronic stability programme system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component repair work / service / replacement carried out, fault code erased, test drive (various road conditions) executed according to manufacturer diagnostic procedure. 4.5 Electronic stability programme system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
5. Diagnose Tyre Pressure Monitoring System (TPMS).	5.1 Review technical investigation and service information report. 5.2 Locate TPMS system fault. 5.3 Troubleshoot TPMS system fault. 5.4 Rectify TPMS system fault. 5.5 Test TPMS system. 5.6 Produce TPMS system diagnostic report.	5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (wiring issue, indicator, signal issue, sensor issue, tyre issue) inspected and electrical components of TPMS system tested according to manufacturer diagnostic procedure. 5.3 Error code identified, TPMS wiring diagram interpreted, malfunction symptom of TPMS detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		5.4 Component service / replacement, programming and calibration (static/dynamic) carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 5.5 TPMS re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

CU TITLE & CU CODE	Perform automotive powertrain system diagnosis. G452-011-4:2025-C04
CU DESCRIPTOR	Perform automotive powertrain system diagnosis describes competency to troubleshoot and resolve issues within the transmission and related control systems that drive the vehicle including fuel injection, ignition, emissions, and transmission systems—as well as the ability to interpret data from diagnostic tools like OBD-II scanners, multimeters, and oscilloscopes. A competent diagnostician can identify and isolate issues accurately, understand how different powertrain components interact, and apply systematic problem-solving. The person who is competent in this CU should be able to diagnose automatic transmission / transaxle electrical system, diagnose Continuous Variable Transmission (CVT) electrical system and diagnose Dual Clutch Transmission (DCT) electrical system. The outcome of this CU is vehicle that shifts smoothly, operates efficiently, and performs reliably under all driving conditions that the transmission functions as intended, with optimal power delivery and fuel efficiency, reducing the likelihood of future breakdowns, extending the lifespan of transmission components and improving the overall driving experience.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
1. Diagnose automatic transmission / transaxle electrical system.	1.1 Review technical investigation and service information report. 1.2 Locate automatic transmission / transaxle mechatronic system fault. 1.3 Troubleshoot automatic transmission / transaxle system fault. 1.4 Rectify automatic transmission / transaxle system fault. 1.5 Test automatic transmission / transaxle system.	1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue) inspected and electrical components of automatic transmission / transaxle system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, automatic transmission / transaxle system wiring diagram interpreted, malfunction symptom of automatic transmission / transaxle system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component repair work / service / replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
	1.6 Produce automatic transmission / transaxle system diagnostic report.	1.5 Automatic transmission / transaxle system functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
2. Diagnose Continuous Variable Transmission (CVT) electrical system.	2.1 Review technical investigation and service information report. 2.2 Locate CVT mechatronic system fault. 2.3 Troubleshoot CVT system fault. 2.4 Rectify CVT system fault. 2.5 Test CVT system. 2.6 Produce CVT system diagnostic report.	2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue) inspected and electrical components of CVT system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, CVT system wiring diagram interpreted, malfunction symptom of CVT system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 2.5 CVT system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
3. Diagnose Dual Clutch Transmission (DCT) electrical system.	3.1 Review technical investigation and service information report. 3.2 Locate DCT mechatronic system fault. 3.3 Troubleshoot DCT electrical system fault. 3.4 Rectify DCT electrical system fault. 3.5 Test DCT electrical system. 3.6 Produce DCT electrical system diagnostic report.	3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue, shift by wire issue) inspected and electrical components of DCT mechatronic system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, DCT electrical system wiring diagram interpreted, malfunction symptom of DCT electrical system detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		3.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 3.5 DCT electrical system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

CU TITLE & CU CODE	Perform automotive Advance Driver Assistance System (ADAS) diagnosis. G452-011-4:2025-C05
CU DESCRIPTOR	Perform automotive Advance Driver Assistance System (ADAS) diagnosis describes competency to accurately assess, troubleshoot, and calibrate complex ADAS components like radar, cameras, sensors, and control modules. It involves understanding how each ADAS feature (such as adaptive cruise control, lane-keeping assist, and automatic emergency braking) functions and integrates with other vehicle systems, as well as knowing how to use specialized diagnostic tools to retrieve and interpret error codes and sensor data. The person who is competent in this CU should be able to diagnose radar system, diagnose camera system, diagnose ultrasonic sensor and diagnose Light Detection and Ranging (LiDAR) system. The outcome of this CU is the system is dependable, responsive, and accurate, leading to improved safety, comfort, and confidence for the driver. It minimizes unnecessary interventions, reduces false alerts, and maximizes the system's operational lifespan, providing a smooth and effective driving experience that integrates seamlessly with other vehicle functions.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
1. Diagnose radar system.	1.1 Review technical investigation and service information report. 1.2 Locate radar system fault. 1.3 Troubleshoot radar system fault. 1.4 Rectify radar system fault. 1.5 Test radar system. 1.6 Produce radar system diagnostic report.	1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms inspected (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue, indicator, system error, cruise control issue, software issue) and electrical components of radar system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, radar system wiring diagram interpreted, malfunction symptom of radar system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 1.5 Radar system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
2 Diagnose camera system.	2.1 Review technical investigation and service information report. 2.2 Locate camera system fault. 2.3 Troubleshoot camera system fault. 2.4 Rectify camera system fault. 2.5 Test camera system. 2.6 Produce camera system diagnostic report.	2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms inspected (monitoring issue, sensing issue, wiring issue, indicator, sensor issue, false warning issue) and electrical components of camera system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, camera system wiring diagram interpreted, malfunction symptom of camera detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 2.5 Camera re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
3 Diagnose ultrasonic sensor system.	3.1 Review technical investigation and service information report. 3.2 Locate ultrasonic sensor system fault. 3.3 Troubleshoot ultrasonic sensor system fault. 3.4 Rectify ultrasonic sensor system fault. 3.5 Test ultrasonic sensor system. 3.6 Produce ultrasonic sensor system diagnostic report.	3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms inspected (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue, indicator, system error, cruise control issue, software issue) and electrical components of ultrasonic sensor system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, ultrasonic sensor system wiring diagram interpreted, malfunction symptom of ultrasonic sensor system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	WORK STEPS	PERFORMANCE CRITERIA
		3.5 Ultrasonic sensor system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.
4 Diagnose Light Detection and Ranging (LiDAR) system.	4.1 Review technical investigation and service information report. 4.2 Locate LiDAR system fault. 4.3 Troubleshoot LiDAR system fault. 4.4 Rectify LiDAR system fault. 4.5 Test LiDAR system. 4.6 Produce LiDAR system diagnostic report.	4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue, indicator, system error, cruise control issue, software issue, physical and mechanical issue) inspected and electrical components of LiDAR system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, LiDAR system wiring diagram interpreted, malfunction symptom of LiDAR system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 4.5 LiDAR system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared according to company policy.

CURRICULUM OF COMPETENCY UNIT

NATIONAL OCCUPATIONAL SKILLS STANDARD (NOSS) FOR:

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

15. Curriculum of Competency Unit

15.1 Perform automotive Engine Management System (EMS) diagnosis.

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
COMPETENCY UNIT TITLE	Perform automotive Engine Management System (EMS) diagnosis.		
LEARNING OUTCOMES	The learning outcomes of this competency are to enable the trainees to investigate, troubleshoot, identify, and resolve issues within the engine's electronic control systems. This includes understanding how EMS components—like fuel injectors, ignition coils, sensors, and the ECM (Engine Control Module) - work together to regulate engine performance. Upon completion of this competency unit, trainees should be able to: 1. Diagnose ignition system. 2. Diagnose starting system. 3. Diagnose charging system. 4. Diagnose electric fuel delivery system. 5. Diagnose engine control system.		
TRAINING PREREQUISITE (SPECIFIC)	Not Available.		
CU CODE	G452-011-4:2025-C01	NOSS LEVEL	Four (4)

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
1. Diagnose ignition system.	1.1 Ignition system safety:	1.1 Review technical investigation and service information report.	<u>ATTITUDE</u> 1.1 Attention to detail while diagnosing ignition system fault.	<u>COGNITIVE DOMAIN</u> 1.1 Ignition system safety explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 1.2 Ignition system technical investigation information: • Vehicle ignition system (types, layout, component, specification). • Types of complaint. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream.	1.2 Locate ignition system fault. 1.3 Troubleshoot ignition system fault. 1.4 Rectify ignition system fault. 1.5 Test ignition system. 1.6 Produce ignition system diagnostic report.	1.2 Systematic troubleshooting ignition system issues. <u>SAFETY</u> 1.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 1.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 1.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 1.1 Proper disposal of hazardous materials in accordance with environmental regulations. 1.2 Minimize emissions and pollution in	1.2 Ignition system technical investigation information described. 1.3 Ignition system diagnostic tools application explained. 1.4 Ignition system fault identification explained. 1.5 Ignition system fault troubleshooting procedure listed. 1.6 Ignition system fault rectification procedure described. 1.7 Ignition system testing procedure identified. 1.8 Type of ignition system diagnostic report explained. 1.9 Ignition system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (no start, misfire, rough idling, wiring issue) inspected and electrical

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Waveform interpretation. • Software version status. 1.3 Ignition system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Ignition coil analyzer. • Emission analyzer. • Compression test kit. • Test lead kit. • Breakout box. • Hand tools. • Soldering tool kit. • Electrical repair tool kit.		accordance with environmental regulations.	components of ignition system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, ignition system wiring diagram interpreted, malfunction symptom of ignition system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 1.5 Ignition system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	1.4 Ignition system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 1.5 Ignition system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 1.6 Ignition system fault rectification procedure: • Rescanning fault code. • Component repair.			AFFECTIVE DOMAIN 1.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 1.2 Step-by-step approach followed, faults identified and resolved systematically. 1.3 Appropriate PPE complied according to safety requirement. 1.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 1.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 1.6 Hazardous materials such as used oil, coolant, batteries, and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Component service. • Component replacement. 1.7 Ignition system testing procedure. • Static test. • Dynamic test. • Functionality test. 1.8 Type of ignition system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure.			other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 1.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	1.9 Ignition system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
2. Diagnose starting system.	2.1 Starting system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling.	2.1 Review technical investigation and service information report. 2.2 Locate starting system fault. 2.3 Troubleshoot starting system fault. 2.4 Rectify starting system fault.	<u>ATTITUDE</u> 2.1 Attention to detail while diagnosing starting system fault. 2.2 Systematic troubleshooting starting system issues.	<u>COGNITIVE DOMAIN</u> 2.1 Starting system safety explained. 2.2 Starting system technical investigation information described. 2.3 Starting system diagnostic tools application explained. 2.4 Starting system fault identification explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.2 Starting system technical investigation information: • Vehicle starting system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 2.3 Starting system diagnostic tools application: • Scan tools.	2.5 Test starting system. 2.6 Produce starting system diagnostic report.	<u>SAFETY</u> 2.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 2.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 2.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 2.1 Proper disposal of hazardous materials in accordance with environmental regulations. 2.2 Minimize emissions and pollution in accordance with environmental regulations.	2.5 Starting system fault troubleshooting procedure listed. 2.6 Starting system fault rectification procedure described 2.7 Starting system testing procedure identified. 2.8 Type of starting system diagnostic report explained. 2.9 Starting system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (cranking issues, wiring issue) inspected and electrical components of starting system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, starting system wiring diagram interpreted, malfunction symptom of starting system

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Digital multimeter. • Oscilloscope with special service tool. • Test lead kit. • Breakout box. • Hand tools. • Soldering tool kit. • Electrical repair tool kit. 2.4 Starting system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 2.5 Starting system fault troubleshooting procedure.			detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 2.5 Starting system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 2.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 2.2 Step-by-step approach followed, faults identified and resolved systematically. 2.3 Appropriate PPE complied according to safety requirement.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Intermittent fault. • Permanent fault. • Unknown fault. 2.6 Starting system fault rectification procedure: • Rescanning fault code. • Component repair. • Component service. • Component replacement. 2.7 Starting system testing procedure. • Static test. • Functionality test. 2.8 Type of starting system diagnostic report: • Summary report.			2.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 2.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 2.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 2.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 2.9 Starting system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation.			

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Approval by superior.			
3. Diagnose charging system.	3.1 Charging system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 3.2 Charging system technical investigation information: • Vehicle charging system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history.	3.1 Review technical investigation and service information report. 3.2 Locate charging system fault. 3.3 Troubleshoot charging system fault. 3.4 Rectify charging system fault. 3.5 Test charging system. 3.6 Produce charging system diagnostic report.	<u>ATTITUDE</u> 3.1 Attention to detail while diagnosing charging system fault. 3.2 Systematic troubleshooting charging system issues. <u>SAFETY</u> 3.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 3.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 3.3 Maintain a safe and organized work area in accordance with safety regulations.	<u>COGNITIVE DOMAIN</u> 3.1 Charging system safety explained. 3.2 Charging system technical investigation information described. 3.3 Charging system diagnostic tools application explained. 3.4 Charging system fault identification explained. 3.5 Charging system fault troubleshooting procedure listed. 3.6 Charging system fault rectification procedure described 3.7 Charging system testing procedure identified. 3.8 Type of charging system diagnostic report explained. 3.9 Charging system diagnostic report format and procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Functionality test result. • Live data stream. • Waveform interpretation. 3.3 Charging system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Ampere Clamp meter. • Battery tester. • Oscilloscope. • Test lead kit. • Breakout box. • Battery charger. • Hand tools. • Soldering tool kit. • Electrical repair tool kit.		<u>ENVIRONMENT</u> 3.1 Proper disposal of hazardous materials in accordance with environmental regulations. 3.2 Minimize emissions and pollution in accordance with environmental regulations.	<u>PSYCHOMOTOR DOMAIN</u> 3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (charging issue – low charge, over charge) inspected and electrical components of charging system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, charging system wiring diagram interpreted, malfunction symptom of charging system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 3.5 Charging system re-diagnosed and its functionality confirmed

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	3.4 Charging system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 3.5 Charging system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 3.6 Charging system fault rectification procedure: • Rescanning fault code. • Component repair.			according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin, warranty report prepared. <u>AFFECTIVE DOMAIN</u> 3.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 3.2 Step-by-step approach followed, faults identified and resolved systematically. 3.3 Appropriate PPE complied according to safety requirement. 3.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 3.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Component service. • Component replacement. 3.7 Charging system testing procedure. • Static test. • Dynamic test. • Functionality test. 3.8 Type of charging system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure.			underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 3.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 3.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	3.9 Charging system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
4. Diagnose electric fuel delivery system.	4.1 Electric fuel delivery system safety: • Usage and application of PPE. • Work area preparation.	4.1 Review technical investigation and service information report. 4.2 Locate electric fuel delivery system fault. 4.3 Troubleshoot electric fuel delivery system fault.	<u>ATTITUDE</u> 4.1 Attention to detail while diagnosing electric fuel delivery system fault. 4.2 Systematic troubleshooting electric fuel delivery system issues.	<u>COGNITIVE DOMAIN</u> 4.1 Electric fuel delivery system safety explained. 4.2 Electric fuel delivery system technical investigation information described. 4.3 Electric fuel delivery system diagnostic tools application explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Tools and equipment handling. 4.2 Electric fuel delivery system technical investigation information: • Vehicle electric fuel delivery system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status.	4.4 Rectify electric fuel delivery system fault. 4.5 Test electric fuel delivery system. 4.6 Produce electric fuel delivery system diagnostic report.	<u>SAFETY</u> 4.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 4.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 4.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 4.1 Proper disposal of hazardous materials in accordance with environmental regulations. 4.2 Minimize emissions and pollution in accordance with environmental regulations.	4.4 Electric fuel delivery system fault identification explained. 4.5 Electric fuel delivery system fault troubleshooting procedure listed. 4.6 Electric fuel delivery system fault rectification procedure described. 4.7 Electric fuel delivery system testing procedure identified. 4.8 Type of electric fuel delivery system diagnostic report explained. 4.9 Electric fuel delivery system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (cranking (starting issue, electric fuel pump issue, fuel injector issue) inspected and electrical components of ignition system tested according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	4.3 Electric fuel delivery system diagnostic tools application: • Fuel pressure gauge. • Injector tester. • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Emission analyzer. • Breakout box. • Test lead kit. • Noid light tester. • Ampere clamp meter. • Hand tools. • Soldering tool kit. • Electrical repair tool kit. 4.4 Electric fuel delivery system			4.3 Error code identified, electric fuel delivery system wiring diagram interpreted, malfunction symptom of electric fuel delivery system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to wiring diagram and manufacturer standard. 4.5 Electric fuel delivery system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 4.5 Electric fuel delivery system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 4.6 Electric fuel delivery system fault rectification procedure: • Rescanning fault code.			<u>AFFECTIVE DOMAIN</u> 4.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 4.2 Step-by-step approach followed, faults identified and resolved systematically. 4.3 Appropriate PPE complied according to safety requirement. 4.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 4.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 4.6 Hazardous materials such as used oil, coolant, batteries, and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Component repair. • Component service. • Component replacement. 4.7 Electric fuel delivery system testing procedure. • Static test. • Dynamic test. • Functionality test. 4.8 Type of electric fuel delivery system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report.			other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 4.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Authority (JPJ) report procedure. 4.9 Electric fuel delivery system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
5. Diagnose engine control system.	5.1 Engine control system safety: • Usage and application of PPE.	5.1 Review technical investigation and service information report. 5.2 Locate engine control system fault.	ATTITUDE 5.1 Attention to detail while diagnosing engine control system fault.	COGNITIVE DOMAIN 5.1 Engine control system safety explained. 5.2 Engine control system technical investigation information described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Work area preparation. • Tools and equipment handling. 5.2 Engine control system technical investigation information: • Vehicle engine control system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status.	5.3 Troubleshoot engine control system fault. 5.4 Rectify engine control system fault. 5.5 Test engine control system. 5.6 Produce engine control system diagnostic report.	5.2 Systematic troubleshooting engine control system issues. <u>SAFETY</u> 5.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 5.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 5.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 5.1 Proper disposal of hazardous materials in accordance with environmental regulations.	5.3 Engine control system diagnostic tools application explained. 5.4 Engine control system fault identification explained. 5.5 Engine control system fault troubleshooting procedure listed. 5.6 Engine control system fault rectification procedure described. 5.7 Engine control system testing procedure identified. 5.8 Type of engine control system diagnostic report explained. 5.9 Engine control system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (malfunction indicator lamp (MIL), starting issue, misfire issue, emission issue) inspected and electrical components of engine control

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	5.3 Engine control system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Test lead kit. • Breakout box. • Sensor / actuator signal simulator. • Hand tools. • Soldering tool kit. • Electrical repair tool kit. 5.4 Engine control system fault identification: • Wiring diagram. • Waveform (sensors and actuators).		5.2 Minimize emissions and pollution in accordance with environmental regulations.	system tested according to manufacturer diagnostic procedure. 5.3 Error code identified, engine control system wiring diagram, signal protocol (Pulse Width Modulation (PWM), Variable Pulse Width VPW, KWP, K-line, L-line) interpreted, malfunction symptom of engine control system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 5.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to wiring diagram and manufacturer standard. 5.5 Engine control system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Manufacturers data specification. 5.5 Engine control system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 5.6 Engine control system fault rectification procedure: • Rescanning fault code. • Component repair. • Component service. • Component replacement. 5.7 Engine control system testing procedure.			<u>AFFECTIVE DOMAIN</u> 5.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 5.2 Step-by-step approach followed, faults identified and resolved systematically. 5.3 Appropriate PPE complied according to safety requirement. 5.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 5.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Static test. • Dynamic test. • Functionality test (sensor and actuator test). 5.8 Type of engine control system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 5.9 Engine control system diagnostic report format and procedure:			5.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 5.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			

Employability Skills

Core Abilities

- Please refer NCS- Core Abilities latest edition.

Social Values & Social Skills

- Please refer Handbook on Social Skills and Social Values in Technical Education and Vocational Training.

References for Learning Material Development

1	Denton, T. (2004). Automobile Electrical and Electronic Systems (3rd ed.). Oxford: Elsevier Butterworth-Heinemann. ISBN 0750662190.
2	Erjavec, J., & Thompson, R. (2019). Automotive Technology: A Systems Approach (7th ed.). Cengage Learning. ISBN 9781337794213.
3	Gilles, T. (2010). Automotive Engines: Diagnosis, Repair, and Rebuilding (6th ed.). Clifton Park, NY: Delmar Pub. ISBN 9781435486416.
4	Halderman, J. (2019). Automotive Electrical and Engine Performance (8th ed.). Pearson: United States. ISBN 9780135224809.
5	Stoakes, G. (2017). Automotive Oscilloscopes: Waveform Analysis. Graham Stoakes: United Kingdom. ISBN 9780992949266.

15.2 Perform automotive body electrical system diagnosis.

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
COMPETENCY UNIT TITLE	Perform automotive body electrical system diagnosis.		
LEARNING OUTCOMES	The learning outcomes of this competency are to enable the trainees to investigate, troubleshoot and resolve electrical issues within the vehicle's chassis systems. It involves understanding electrical principles, interpreting wiring diagrams, and using diagnostic tools like multimeter and scan tools to pinpoint faults accurately. Upon completion of this competency unit, trainees should be able to: 1. Diagnose lighting and instrument panel system. 2. Diagnose washer and wiper system. 3. Diagnose power window and mirror system. 4. Diagnose infotainment system. 5. Diagnose security and lock system. 6. Diagnose Supplemental Restraint System (SRS). 7. Diagnose Electrical Heating, Ventilation and Air Conditioning system (HVAC).		
TRAINING PREREQUISITE (SPECIFIC)	Not Available.		
CU CODE	G452-011-4:2025-C02	NOSS LEVEL	Four (4)

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
1. Diagnose lighting and	1.1 Lighting and instrument panel system safety:	1.1 Review technical investigation and service	ATTITUDE 1.1 Attention to detail while diagnosing	COGNITIVE DOMAIN 1.1 Lighting and instrument panel system safety explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
instrument panel system.	• Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 1.2 Lighting and instrument panel system technical investigation information: • Vehicle lighting and instrument panel system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result.	information report. 1.2 Locate lighting and instrument panel system fault. 1.3 Troubleshoot lighting and instrument panel system fault. 1.4 Rectify lighting and instrument panel system fault. 1.5 Test lighting and instrument panel system. 1.6 Produce lighting and instrument panel system diagnostic report.	lighting and instrument panel system fault. 1.2 Systematic troubleshooting of lighting and instrument panel system issues. <u>SAFETY</u> 1.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 1.1 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 1.2 Maintain a safe and organized work area in accordance with safety regulations.	1.2 Lighting and instrument panel system technical investigation information described. 1.3 Lighting and instrument panel system diagnostic tools application explained. 1.4 Lighting and instrument panel system fault identification explained. 1.5 Lighting and instrument panel system fault troubleshooting procedure listed. 1.6 Lighting and instrument panel system fault rectification procedure described. 1.7 Lighting and instrument panel system testing procedure identified. 1.8 Type of lighting and instrument panel system diagnostic report explained. 1.9 Lighting and instrument panel system diagnostic report format and procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Software version status. 1.3 Lighting and instrument panel system diagnostic tools application: • Scan tools. • Digital multimeter. • Headlight alignment machine. • Test lead kit. • Breakout box. • Hand tools. 1.4 Lighting and instrument panel system fault identification: • Wiring diagram. • Sensors and actuators test. • Manufacturers data specification. 1.5 Lighting and instrument panel		ENVIRONMENT 1.1 Proper disposal of hazardous materials in accordance with environmental regulations. 1.2 Minimize emissions and pollution in accordance with environmental regulations.	PSYCHOMOTOR DOMAIN 1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (wiring issue, lighting issue, indicator issue, switch issue, body control module issue) inspected and electrical components of lighting and instrument panel system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, lighting and instrument panel system wiring diagram interpreted, malfunction symptom of lighting and instrument panel system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component repair work / service / replacement and programming carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 1.6 Lighting and instrument panel system fault rectification procedure: • Rescanning fault code. • Component replacement. 1.7 Lighting and instrument panel system testing procedure. • Static test. • Dynamic test. • Functionality test (calibration test).			1.5 Lighting and instrument panel system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 1.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 1.2 Step-by-step approach followed, faults identified and resolved systematically. 1.3 Appropriate PPE complied according to safety requirement. 1.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 1.5 Workspace clean and free from clutter to reduce tripping

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	1.8 Type of lighting and instrument panel system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 1.9 Lighting and instrument panel system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result.			hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 1.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 1.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Result analysis. • Justification. • Recommendation. • Approval by superior.			
2. Diagnose washer and wiper system.	2.1 Washer and wiper system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 2.2 Washer and wiper system technical investigation information: • Vehicle washer and wiper system (types, layout, component, specification).	2.1 Review technical investigation and service information report. 2.2 Locate washer and wiper system fault. 2.3 Troubleshoot washer and wiper system fault. 2.4 Rectify washer and wiper system fault. 2.5 Test washer and wiper system. 2.6 Produce washer and wiper system diagnostic report.	<u>ATTITUDE</u> 2.1 Attention to detail while diagnosing washer and wiper system fault. 2.2 Systematic troubleshooting washer and wiper system issues. <u>SAFETY</u> 2.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 2.2 Proper handling of tools and equipment in accordance with safety regulations and	<u>COGNITIVE DOMAIN</u> 2.1 Washer and wiper system safety explained. 2.2 Washer and wiper system technical investigation information described. 2.3 Washer and wiper system diagnostic tools application explained. 2.4 Washer and wiper system fault identification explained. 2.5 Washer and wiper system fault troubleshooting procedure listed. 2.6 Washer and wiper system fault rectification procedure described. 2.7 Washer and wiper system testing procedure identified.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Software version status. 2.3 Washer and wiper system diagnostic tools application: • Scan tools. • Digital multimeter. • Test lead kit. • Hand tools. 2.4 Washer and wiper system fault identification: • Wiring diagram. • Sensors and actuators test. • Manufacturers data specification.		manufacturer requirement. 2.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 2.1 Proper disposal of hazardous materials in accordance with environmental regulations. 2.2 Minimize emissions and pollution in accordance with environmental regulations.	2.8 Type of washer and wiper system diagnostic report explained. 2.9 Washer and wiper system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (wiring issue, motor issue, body control module issue) inspected and electrical components of washer and wiper system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, washer and wiper system wiring diagram interpreted, malfunction symptom of washer and wiper system detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.5 Washer and wiper system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 2.6 Washer and wiper system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 2.7 Washer and wiper system testing procedure. • Static test. • Dynamic test. • Functionality test.			2.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 2.5 Washer and wiper system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 2.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 2.2 Step-by-step approach followed, faults identified and resolved systematically. 2.3 Appropriate PPE complied according to safety requirement. 2.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.8 Type of washer and wiper system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 2.9 Washer and wiper system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis.			working on electrical systems according to manufacturer manual. 2.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 2.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 2.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Justification. • Recommendation. • Approval by superior.			
3. Diagnose power window and mirror system.	3.1 Power window and mirror system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 3.2 Power window and mirror system technical investigation information: • Vehicle power window and mirror system (types, layout, component, specification).	3.1 Review technical investigation and service information report. 3.2 Locate power window and mirror system fault. 3.3 Troubleshoot power window and mirror system fault. 3.4 Rectify power window and mirror system fault. 3.5 Test power window and mirror system. 3.6 Produce power window and mirror system diagnostic report.	<u>ATTITUDE</u> 3.1 Attention to detail while diagnosing power window and mirror system fault. 3.2 Systematic troubleshooting power window and mirror system issues. <u>SAFETY</u> 3.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 3.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 3.3 Maintain a safe and organized work area	<u>COGNITIVE DOMAIN</u> 3.1 Power window and mirror system safety explained. 3.2 Power window and mirror system technical investigation information described. 3.3 Power window and mirror system diagnostic tools application explained. 3.4 Power window and mirror system fault identification explained. 3.5 Power window and mirror system fault troubleshooting procedure listed. 3.6 Power window and mirror system fault rectification procedure described. 3.7 Power window and mirror system testing procedure identified. 3.8 Type of power window and mirror system diagnostic report explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Software version status. 3.3 Power window and mirror system diagnostic tools application: • Scan tools. • Digital multimeter. • Test lead kit. • Hand tools. 3.4 Power window and mirror system fault identification: • Wiring diagram. • Sensors and actuators test.		in accordance with safety regulations. <u>ENVIRONMENT</u> 3.1 Proper disposal of hazardous materials in accordance with environmental regulations. 3.2 Minimize emissions and pollution in accordance with environmental regulations.	3.9 Power window and mirror system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (wiring issue, motor issue, body control module issue, rear demister) inspected and electrical components of power window and mirror system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, power window and mirror system wiring diagram interpreted, malfunction symptom of power window and mirror system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Manufacturers data specification. 3.5 Power window and mirror system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 3.6 Power window and mirror system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 3.7 Power window and mirror system testing procedure. • Static test.			programming carried out; fault code erased according to manufacturer diagnostic procedure. 3.5 Power window and mirror system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 3.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 3.2 Step-by-step approach followed, faults identified and resolved systematically. 3.3 Appropriate PPE complied according to safety requirement. 3.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Dynamic test. • Functionality test. 3.8 Type of window and mirror system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 3.9 Power window and mirror system diagnostic report format and procedure: • Type of report. • Type of fault.			according to manufacturer manual. 3.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 3.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 3.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
4. Diagnose infotainment system.	4.1 Infotainment system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 4.2 Infotainment system technical investigation information: • Vehicle infotainment system (types,	4.1 Review technical investigation and service information report. 4.2 Locate infotainment system fault. 4.3 Troubleshoot infotainment system fault. 4.4 Rectify infotainment system fault. 4.5 Test infotainment system.	<u>ATTITUDE</u> 4.1 Attention to detail while diagnosing infotainment system fault. 4.2 Systematic troubleshooting infotainment system issues. <u>SAFETY</u> 4.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations.	<u>COGNITIVE DOMAIN</u> 4.1 Infotainment system safety explained. 4.2 Infotainment system technical investigation information described. 4.3 Infotainment system diagnostic tools application explained. 4.4 Infotainment system fault identification explained. 4.5 Infotainment system fault troubleshooting procedure listed. 4.6 Infotainment system fault rectification procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Software version status. 4.3 Infotainment system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Breakout box. • Test lead kit. • Hand tools. 4.4 Infotainment system fault identification:	4.6 Produce infotainment system diagnostic report.	4.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 4.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 4.1 Proper disposal of hazardous materials in accordance with environmental regulations. 4.2 Minimize emissions and pollution in accordance with environmental regulations.	4.7 Infotainment system testing procedure identified. 4.8 Type of infotainment system diagnostic report explained. 4.9 Infotainment system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, software issue, signal issue, display issue, connectivity issue) inspected and electrical components of infotainment system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, infotainment system wiring diagram interpreted, malfunction symptom of infotainment system detected and data stream analysis interpreted according to

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Wiring diagram. • Sensors and actuators test. • Manufacturers data specification. 4.5 Infotainment system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 4.6 Infotainment system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement.			manufacturer diagnostic procedure. 4.4 Component service work / replacement and programming carried out; fault code erased according to manufacturer diagnostic procedure. 4.5 Infotainment system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 4.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 4.2 Step-by-step approach followed, faults identified and resolved systematically. 4.3 Appropriate PPE complied according to safety requirement. 4.4 Good working condition tools and diagnostic equipment used

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	4.7 Infotainment system testing procedure. • Static test. • Dynamic test. • Functionality test. 4.8 Type of infotainment system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 4.9 Infotainment system diagnostic report format and procedure:			correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 4.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 4.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 4.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
5. Diagnose security and lock system.	5.1 Security and lock system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 5.2 Security and lock system technical investigation information:	5.1 Review technical investigation and service information report. 5.2 Locate security and lock system fault. 5.3 Troubleshoot security and lock system fault. 5.4 Rectify security and lock system fault.	<u>ATTITUDE</u> 5.1 Attention to detail while diagnosing security and lock system fault. 5.2 Systematic troubleshooting security and lock system issues. <u>SAFETY</u> 5.1 Use of Personal Protective Equipment (PPE) in accordance	<u>COGNITIVE DOMAIN</u> 5.1 Security and lock system safety explained. 5.2 Security and lock system technical investigation information described. 5.3 Security and lock system diagnostic tools application explained. 5.4 Security and lock system fault identification explained. 5.5 Security and lock system fault troubleshooting procedure listed.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Vehicle security and lock system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Software version status. 5.3 Security and lock system diagnostic tools application: • Scan tools. • Digital multimeter. • Test lead kit. • Key transponder detector kit. • Hand tools.	5.5 Test security and lock system. 5.6 Produce security and lock system diagnostic report.	with safety regulations. 5.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 5.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 5.1 Proper disposal of hazardous materials in accordance with environmental regulations. 5.2 Minimize emissions and pollution in accordance with environmental regulations.	5.6 Security and lock system fault rectification procedure described. 5.7 Security and lock system testing procedure identified. 5.8 Type of security and lock system diagnostic report explained. 5.9 Security and lock system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (starting issue, communication issue, wiring issue, alarm issue) inspected and electrical components of security and lock system tested according to manufacturer diagnostic procedure. 5.3 Error code identified, security and lock system wiring diagram interpreted, malfunction symptom of security and lock system

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	5.4 Security and lock system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 5.5 Security and lock system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 5.6 Security and lock system fault rectification procedure: • Rescanning fault code.			detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 5.4 Component repair work / service / replacement carried out; fault code erased according to manufacturer diagnostic procedure. 5.5 Security and lock system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. AFFECTIVE DOMAIN 5.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 5.2 Step-by-step approach followed, faults identified and resolved systematically. 5.3 Appropriate PPE complied according to safety requirement.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Component service. • Component replacement. 5.7 Security and lock system testing procedure. • Static test. • Dynamic test. • Functionality test. 5.8 Type of security and lock system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure.			5.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 5.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 5.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 5.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	5.9 Security and lock system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
6. Diagnose Supplemental Restraint System (SRS).	6.1 SRS safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling.	6.1 Review technical investigation and service information report. 6.2 Locate supplemental restraint system fault. 6.3 Troubleshoot supplemental	<u>ATTITUDE</u> 6.1 Attention to detail while diagnosing SRS fault. 6.2 Systematic troubleshooting SRS issues. <u>SAFETY</u> 6.1 Use of Personal Protective Equipment	<u>COGNITIVE DOMAIN</u> 6.1 SRS safety explained. 6.2 SRS technical investigation information described. 6.3 SRS diagnostic tools explained. 6.4 SRS fault identification explained. 6.5 SRS fault troubleshooting procedure listed.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	6.2 SRS technical investigation information: • Vehicle SRS (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Software version status. 6.3 SRS diagnostic tools application: • Scan tools. • Digital multimeter. • SRS test kit. • Hand tools. 6.4 SRS fault identification:	restraint system fault. 6.4 Rectify supplemental restraint system fault. 6.5 Test supplemental restraint system. 6.6 Produce supplemental restraint system diagnostic report.	(PPE) in accordance with safety regulations. 6.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 6.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 6.1 Proper disposal of hazardous materials in accordance with environmental regulations. 6.2 Minimize emissions and pollution in accordance with environmental regulations.	6.6 SRS fault rectification procedure described. 6.7 SRS testing procedure identified. 6.8 Type of SRS diagnostic report explained. 6.9 SRS diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 6.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 6.2 Symptoms (wiring issue, warning light, sensor issue) inspected and electrical components of supplemental restraint system tested according to manufacturer diagnostic procedure. 6.3 Error code identified, supplemental restraint system wiring diagram interpreted, malfunction symptom of supplemental restraint system detected and data stream analysis interpreted according

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Wiring diagram. • Sensors test. • Manufacturers data specification. 6.5 SRS fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 6.6 SRS fault rectification procedure: • Rescanning fault code. • Component replacement. 6.7 SRS testing procedure. • Static test. • Dynamic test. • Functionality test (seatbelt			to manufacturer diagnostic procedure. 6.4 Component replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 6.5 supplemental restraint system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 6.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 6.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 6.2 Step-by-step approach followed, faults identified and resolved systematically. 6.3 Appropriate PPE complied according to safety requirement.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	buckle, clock spring). 6.8 Type of SRS diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 6.9 SRS diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis.			6.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 6.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 6.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 6.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Justification. • Recommendation. • Approval by superior.			
7. Diagnose Electrical Heating, Ventilation and Air Conditioning system (HVAC).	7.1 HVAC system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. • Low / high voltage. 7.2 HVAC system technical investigation information: • Vehicle HVAC system (types, layout, component, specification). • Type of complaints.	7.1 Review technical investigation and service information report. 7.2 Locate HVAC system fault. 7.3 Troubleshoot ignition system fault. 7.4 Rectify HVAC system fault. 7.5 Test HVAC system. 7.6 Produce HVAC system diagnostic report.	<u>ATTITUDE</u> 7.1 Attention to detail while diagnosing HVAC system fault. 7.2 Systematic troubleshooting HVAC system issues. <u>SAFETY</u> 7.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 7.2 Proper handling of tools and equipment including high voltage HVAC system in accordance with safety regulations and manufacturer requirement. 7.3 Maintain a safe and organized work area	<u>COGNITIVE DOMAIN</u> 7.1 HVAC system safety explained. 7.2 HVAC system technical investigation information described. 7.3 HVAC system diagnostic tools application explained. 7.4 HVAC system fault identification explained. 7.5 HVAC system fault troubleshooting procedure listed. 7.6 HVAC system fault rectification procedure described. 7.7 HVAC system testing procedure identified. 7.8 Type of HVAC system diagnostic report explained. 7.9 HVAC system diagnostic report format and procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 7.3 HVAC system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Air conditioning recovery machine. • Test lead kit. • Manifold gauge.		in accordance with safety regulations. <u>ENVIRONMENT</u> 7.1 Proper disposal of hazardous materials in accordance with environmental regulations. 7.2 Minimize emissions and pollution in accordance with environmental regulations.	<u>PSYCHOMOTOR DOMAIN</u> 7.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 7.2 Symptoms (wiring issue, sensor issue, motor issue, compressor issue, climate control panel issue) inspected and electrical components of HVAC system tested according to manufacturer diagnostic procedure. 7.3 Error code identified, HVAC system wiring diagram interpreted, malfunction symptom of HVAC system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 7.4 Component repair work / service / replacement and programming carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 7.5 HVAC system re-diagnosed and its functionality confirmed

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Refrigerant analyzer. • Wind speed meter. • Digital thermometer. • Leak test kit (UV light, adapter, UV glass, electronic leak detector). • Consumables (refrigerant, compressor oil, UV liquid). • AC disconnection tool set. • AC compressor clutch remover. • Electrical expansion valve test kit. • Hand tools.			according to manufacturer diagnostic procedure. 7.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 7.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 7.2 Step-by-step approach followed, faults identified and resolved systematically. 7.3 Appropriate PPE complied according to safety requirement. 7.4 Good working condition tools and diagnostic equipment used correctly (including high voltage HVAC system), and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 7.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Soldering tool kit. • Electrical repair tool kit. 7.4 HVAC system fault identification: • Wiring diagram. • Sensors and actuators test. • Pressure test. • Vacuum test. • Manufacturers data specification. 7.5 HVAC system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault.			vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 7.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 7.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	7.6 HVAC system fault rectification procedure: • Rescanning fault code. • Component repair. • Component service. • Component replacement. 7.7 HVAC system testing procedure. • Static test. • Dynamic test. • Functionality test. 7.8 Type of HVAC system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign.			

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Warranty report. • Authority (JPJ) report procedure. 7.9 HVAC system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			

Employability Skills

Core Abilities

- Please refer NCS- Core Abilities latest edition.

Social Values & Social Skills

- Please refer Handbook on Social Skills and Social Values in Technical Education and Vocational Training.

References for Learning Material Development

1	Asef, P., Padmanaban, S., & Lapthorn, A. (Eds.) (2022). Modern Automotive Electrical Systems. Wiley. ISBN 9781119801078.
2	Halderman, J. (2016). Automotive Electricity and Electronics (5th ed.). Upper Saddle River, NJ: Pearson. ISBN 9780134073644.
3	Haynes, A. (2006). Electrical and Electronic Systems (1st ed.). Somerset: Haynes Publishing. ISBN 9781844252510.
4	Rajoli, N. M. (2008). Sistem Penyamanan Udara Kenderaan. Petaling Jaya, Selangor: IBS Buku. ISBN 9789679502701.
5	Stoakes, G. (2017). Automotive Oscilloscopes: Waveform Analysis. Graham Stoakes: United Kingdom. ISBN 9780992949266.
6	White, C. (1998). Automotive Diagnostic Fault Codes Manual (1st ed.). Somerset: Haynes Manuals Inc. ISBN 9781859604724.

15.3 Perform automotive chassis electrical diagnosis.

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
COMPETENCY UNIT TITLE	Perform automotive chassis electrical diagnosis.		
LEARNING OUTCOMES	The learning outcomes of this competency are to enable the trainees to investigate, troubleshoot and resolve electrical issues within the vehicle's chassis systems. It involves understanding electrical principles, interpreting wiring diagrams, and using diagnostic tools like multimeters and scan tools to pinpoint faults accurately. Upon completion of this competency unit, trainees should be able to: 1. Diagnose Electric Power Steering (EPS) system. 2. Diagnose Anti-lock Braking System (ABS). 3. Diagnose Electronic Parking Brake (EPB). 4. Diagnose Electronic Stability Programme (ESP) system. 5. Diagnose Tyre Pressure Monitoring System (TPMS).		
TRAINING PREREQUISITE (SPECIFIC)	Not Available.		
CU CODE	G452-011-4:2025-C03	NOSS LEVEL	Four (4)

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
1. Diagnose Electric Power Steering	1.1 EPS system safety: • Usage and application of PPE.	1.1 Review technical investigation and service information report.	<u>ATTITUDE</u> 1.1 Attention to detail while diagnosing EPS system fault.	<u>COGNITIVE DOMAIN</u> 1.1 EPS system safety explained. 1.2 EPS system technical investigation information described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
(EPS) system.	• Work area preparation. • Tools and equipment handling. 1.2 EPS system technical investigation information: • Vehicle EPS system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Waveform. • Live data stream. • Software version status.	1.2 Locate electric power steering system fault. 1.3 Troubleshoot electric power steering system fault. 1.4 Rectify electric power steering system fault. 1.5 Test electric power steering system. 1.6 Produce electric power steering system diagnostic report.	1.2 Systematic troubleshooting EPS system issues. <u>SAFETY</u> 1.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 1.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 1.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 1.1 Proper disposal of hazardous materials in accordance with environmental regulations. 1.2 Minimize emissions and pollution in	1.3 EPS system diagnostic tools application explained. 1.4 EPS system fault identification explained. 1.5 EPS system fault troubleshooting procedure listed. 1.6 EPS system fault rectification procedure described. 1.7 EPS system testing procedure identified. 1.8 Type of EPS system diagnostic report explained. 1.9 EPS system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (steering issue, starting issue, wiring issue, indicator, sensor) inspected and electrical components of electric power steering system tested according to

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	1.3 EPS system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • AC/DC clamp meter. • Test lead kit. • Hand tools. • Wheel alignment machine. 1.4 EPS system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification.		accordance with environmental regulations.	manufacturer diagnostic procedure. 1.3 Error code identified, electric power steering system wiring diagram interpreted, malfunction symptom of electric power steering system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 1.5 Electric power steering system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	1.5 EPS system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 1.6 EPS system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 1.7 EPS system testing procedure. • Static test. • Dynamic test. • Functionality test • Calibration test (wheel alignment,			<u>AFFECTIVE DOMAIN</u> 1.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 1.2 Step-by-step approach followed, faults identified and resolved systematically. 1.3 Appropriate PPE complied according to safety requirement. 1.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 1.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 1.6 Hazardous materials such as used oil, coolant, batteries, and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	motor calibration, steering angle). 1.8 Type of EPS system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 1.9 EPS system diagnostic report format and procedure: • Type of report. • Type of fault.			other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 1.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
2. Diagnose Anti-lock Braking System (ABS).	2.1 ABS safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 2.2 ABS technical investigation information: • Vehicle ABS (types, layout, component, specification).	2.1 Review technical investigation and service information report. 2.2 Locate anti-lock brake system fault. 2.3 Troubleshoot anti-lock brake system fault. 2.4 Rectify anti-lock brake system fault. 2.5 Test anti-lock brake system. 2.6 Produce anti-lock brake system diagnostic report.	<u>ATTITUDE</u> 2.1 Attention to detail while diagnosing ABS fault. 2.2 Systematic troubleshooting ABS issues. <u>SAFETY</u> 2.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 2.2 Proper handling of tools and equipment in accordance with	<u>COGNITIVE DOMAIN</u> 2.1 ABS safety explained. 2.2 ABS technical investigation information described. 2.3 ABS diagnostic tools explained. 2.4 ABS fault identification explained. 2.5 ABS fault troubleshooting procedure listed. 2.6 ABS fault rectification procedure described. 2.7 ABS testing procedure identified. 2.8 Type of ABS diagnostic report explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 2.3 ABS diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Breakout box. • Test lead kit. • Brake bleeding machine. • Hand tools.		safety regulations and manufacturer requirement. 2.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 2.1 Proper disposal of hazardous materials in accordance with environmental regulations. 2.2 Minimize emissions and pollution in accordance with environmental regulations.	2.9 ABS diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (wiring issue, indicator, brake issue, sensor issue) inspected and electrical components of anti-lock brake system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, Anti-lock brake system wiring diagram interpreted, malfunction symptom of Anti-lock brake system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component repair work / service / replacement, programming and calibration carried out, fault code erased, test drive executed according to

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.4 ABS fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 2.5 ABS fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 2.6 ABS fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement.			manufacturer diagnostic procedure. 2.5 Anti-lock brake system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 2.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 2.2 Step-by-step approach followed, faults identified and resolved systematically. 2.3 Appropriate PPE complied according to safety requirement. 2.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.7 ABS testing procedure. • Static test. • Dynamic test. • Functionality test. 2.8 Type of ABS diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 2.9 ABS diagnostic report format and procedure: • Type of report. • Type of fault.			2.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 2.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 2.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
3. Diagnose Electronic Parking Brake (EPB) system.	3.1 EPB system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 3.2 EPB system technical investigation information: • Vehicle EPB system (types, layout,	3.1 Review technical investigation and service information report. 3.2 Locate EPB system fault. 3.3 Troubleshoot ignition system fault. 3.4 Rectify EPB system fault. 3.5 Test EPB system. 3.6 Produce EPB system diagnostic report.	ATTITUDE 3.1 Attention to detail while diagnosing EPB system fault. 3.2 Systematic troubleshooting EPB system issues. SAFETY 3.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 3.2 Proper handling of tools and equipment in accordance with	COGNITIVE DOMAIN 3.1 EPB system safety explained. 3.2 EPB system technical investigation information described. 3.3 EPB system diagnostic tools application explained. 3.4 EPB system fault identification explained. 3.5 EPB system fault troubleshooting procedure listed. 3.6 EPB system fault rectification procedure described. 3.7 EPB system testing procedure identified.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 3.3 EPB system diagnostic tools application: • Scan tools. • Digital multimeter. • Test lead kit. • Piston brake retractor kit. • Hand tools. 3.4 EPB system fault identification:		safety regulations and manufacturer requirement. 3.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 3.1 Proper disposal of hazardous materials in accordance with environmental regulations. 3.2 Minimize emissions and pollution in accordance with environmental regulations.	3.8 Type of EPB system diagnostic report explained. 3.9 EPB system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (wiring issue, indicator, brake issue, sensor issue, motor issue, switch issue) inspected and electrical components of EPB system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, EPB system wiring diagram interpreted, malfunction symptom of EPB detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement, programming and calibration

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Wiring diagram. • Sensors and actuators test. • Manufacturers data specification. 3.5 EPB system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 3.6 EPB system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 3.7 EPB system testing procedure.			carried out; fault code erased according to manufacturer diagnostic procedure. 3.5 EPB system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure.\\ 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 3.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 3.2 Step-by-step approach followed, faults identified and resolved systematically. 3.3 Appropriate PPE complied according to safety requirement. 3.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Static test. • Functionality test. 3.8 Type of EPB system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 3.9 EPB system diagnostic report format and procedure: • Type of report. • Type of fault.			3.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 3.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 3.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
4. Diagnose Electronic Stability Programme (ESP) system.	4.1 ESP system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 4.2 ESP system technical investigation information: • Vehicle ESP system (types, layout,	4.1 Review technical investigation and service information report. 4.2 Locate electronic stability programme system fault. 4.3 Troubleshoot ignition system fault. 4.4 Rectify electronic stability programme system fault. 4.5 Test electronic stability programme system. 4.6 Produce electronic stability programme	<u>ATTITUDE</u> 4.1 Attention to detail while diagnosing ESP system fault. 4.2 Systematic troubleshooting ESP system issues. <u>SAFETY</u> 4.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 4.2 Proper handling of tools and equipment in accordance with	<u>COGNITIVE DOMAIN</u> 4.1 ESP system safety explained. 4.2 ESP system technical investigation information described. 4.3 ESP system diagnostic tools application explained. 4.4 ESP system fault identification explained. 4.5 ESP system fault troubleshooting procedure listed. 4.6 ESP system fault rectification procedure described. 4.7 ESP system testing procedure identified.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 4.3 ESP system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Breakout box. • Test lead kit.	system diagnostic report.	safety regulations and manufacturer requirement. 4.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 4.1 Proper disposal of hazardous materials in accordance with environmental regulations. 4.2 Minimize emissions and pollution in accordance with environmental regulations.	4.8 Type of ESP system diagnostic report explained. 4.9 ESP system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, indicator, brake issue, steering angle sensor issue, sensor issue) inspected and electrical components of electronic stability programme system tested according to manufacturer diagnostic procedure. 4.3 Error code identified, electronic stability programme system wiring diagram interpreted, malfunction symptom of electronic stability programme system detected and data stream analysis interpreted according to

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Brake bleeding machine. • Hand tools. 4.4 ESP system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 4.5 ESP system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 4.6 ESP system fault rectification procedure: • Rescanning fault code.			manufacturer diagnostic procedure. 4.4 Component repair work / service / replacement carried out, fault code erased, test drive (various road conditions) executed according to manufacturer diagnostic procedure. 4.5 Electronic stability programme system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 4.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 4.2 Step-by-step approach followed, faults identified and resolved systematically. 4.3 Appropriate PPE complied according to safety requirement.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Component service. • Component replacement. 4.7 ESP system testing procedure. • Static test. • Dynamic test. • Functionality test. 4.8 Type of ESP system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure.			4.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 4.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 4.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 4.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	4.9 ESP system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
5. Diagnose Tyre Pressure Monitoring System (TPMS).	5.1 TPMS safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling.	5.1 Review technical investigation and service information report. 5.2 Locate TPMS system fault. 5.3 Troubleshoot TPMS system fault. 5.4 Rectify TPMS system fault.	<u>ATTITUDE</u> 5.1 Attention to detail while diagnosing TPMS fault. 5.2 Systematic troubleshooting TPMS system issues.	<u>COGNITIVE DOMAIN</u> 5.1 TPMS safety explained. 5.2 TPMS technical investigation information described. 5.3 TPMS diagnostic tools explained. 5.4 TPMS fault identification explained. 5.5 TPMS fault troubleshooting procedure listed.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	5.2 TPMS technical investigation information: • Vehicle TPMS (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Frequency interpretation. • Software version status. 5.3 TPMS diagnostic tools application: • Scan tools. • TPMS diagnostic tools.	5.5 Test TPMS system. 5.6 Produce TPMS system diagnostic report.	<u>SAFETY</u> 5.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 5.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 5.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 5.1 Proper disposal of hazardous materials in accordance with environmental regulations. 5.2 Minimize emissions and pollution in accordance with environmental regulations.	5.6 TPMS fault rectification procedure described. 5.7 TPMS testing procedure identified. 5.8 Type of TPMS diagnostic report explained. 5.9 TPMS diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 5.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 5.2 Symptoms (wiring issue, indicator, signal issue, sensor issue, tyre issue) inspected and electrical components of TPMS system tested according to manufacturer diagnostic procedure. 5.3 Error code identified, TPMS wiring diagram interpreted, malfunction symptom of TPMS detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Digital multimeter. • Tyre pressure gauge. • Tyre changer. • Hand tools. 5.4 TPMS fault identification: • Wiring diagram. • Frequency (sensors and actuators). • Manufacturers data specification. 5.5 TPMS fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 5.6 TPMS fault rectification procedure:			5.4 Component service / replacement, programming and calibration (static/dynamic) carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 5.5 TPMS re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 5.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. AFFECTIVE DOMAIN 5.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 5.2 Step-by-step approach followed, faults identified and resolved systematically. 5.3 Appropriate PPE complied according to safety requirement. 5.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Rescanning fault code. • Component replacement. 5.7 TPMS testing procedure. • Static test. • Dynamic test. • Functionality test. 5.8 Type of TPMS diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 5.9 TPMS diagnostic report format and procedure:			battery disconnected when working on electrical systems according to manufacturer manual. 5.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 5.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 5.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			

Employability Skills

Core Abilities

- Please refer NCS- Core Abilities latest edition.

Social Values & Social Skills

- Please refer Handbook on Social Skills and Social Values in Technical Education and Vocational Training.

References for Learning Material Development

1	Halderman, J. (2016). Automotive Electricity and Electronics (5th ed.). Upper Saddle River, NJ: Pearson. ISBN 9780134073644.
2	Asef, P., Padmanaban, S., & Lapthorn, A. (Eds.) (2022). Modern Automotive Electrical Systems. Wiley. ISBN 9781119801078.
3	White, C. (1998). Automotive Diagnostic Fault Codes Manual (1st ed.). Somerset: Haynes Manuals Inc. ISBN 9781859604724.
4	Haynes, A. (2006). Electrical and Electronic Systems (1st ed.). Somerset: Haynes Publishing. ISBN 9781844252510.
5	Stoakes, G. (2017). Automotive Oscilloscopes: Waveform Analysis. Graham Stoakes: United Kingdom. ISBN 9780992949266.
6	Crouse, W. H. (1993). Automotive Mechanics (10th ed.). Career Education: New York. ISBN 9780028009438.

15.4 Perform automotive powertrain system diagnosis.

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
COMPETENCY UNIT TITLE	Perform automotive powertrain system diagnosis.		
LEARNING OUTCOMES	The learning outcomes of this competency are to enable the trainees to investigate, troubleshoot and resolve issues within the transmission and related control systems that drive the vehicle including fuel injection, ignition, emissions, and transmission systems—as well as the ability to interpret data from diagnostic tools like OBD-II scanners, multimeters, and oscilloscopes. Upon completion of this competency unit, trainees should be able to: 1. Diagnose automatic transmission / transaxle electrical system. 2. Diagnose Continuous Variable Transmission (CVT) electrical system. 3. Diagnose Dual Clutch Transmission (DCT) electrical system.		
TRAINING PREREQUISITE (SPECIFIC)	Not Available.		
CU CODE	G452-011-4:2025-C04	NOSS LEVEL	Four (4)

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
1. Diagnose automatic transmission / transaxle electrical system.	1.1 Automatic transmission / transaxle electrical system safety:	1.1 Review technical investigation and service information report. 1.2 Locate automatic transmission /	<u>ATTITUDE</u> 1.1 Attention to detail while diagnosing automatic transmission / transaxle electrical system fault.	<u>COGNITIVE DOMAIN</u> 1.1 Automatic transmission / transaxle electrical system safety explained. 1.2 Transmission / transaxle electrical system technical

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 1.2 Automatic transmission / transaxle electrical system technical investigation information: • Vehicle automatic transmission / transaxle electrical system (layout, component, specification). • Type of complaints. • Fault symptoms.	transaxle mechatronic system fault. 1.3 Troubleshoot automatic transmission / transaxle system fault. 1.4 Rectify automatic transmission / transaxle system fault. 1.5 Test automatic transmission / transaxle system. 1.6 Produce automatic transmission / transaxle system diagnostic report.	1.2 Systematic troubleshooting automatic transmission / transaxle electrical system issues. <u>SAFETY</u> 1.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 1.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 1.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 1.1 Proper disposal of hazardous materials in accordance with	investigation information described. 1.3 Transmission / transaxle electrical system diagnostic tools application explained. 1.4 Transmission / transaxle electrical system fault identification explained. 1.5 Transmission / transaxle electrical system fault troubleshooting procedure listed. 1.6 Transmission / transaxle electrical system fault rectification procedure described. 1.7 Transmission / transaxle electrical system testing procedure identified. 1.8 Type of transmission / transaxle electrical system diagnostic report explained. 1.9 Transmission / transaxle electrical system diagnostic report format and procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 1.3 Automatic transmission / transaxle electrical system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Hand tools. • Test lead kit. • Oil pressure gauge. • Soldering tool kit.		environmental regulations. 1.2 Minimize emissions and pollution in accordance with environmental regulations.	<u>PSYCHOMOTOR DOMAIN</u> 1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue) inspected and electrical components of automatic transmission / transaxle system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, automatic transmission / transaxle system wiring diagram interpreted, malfunction symptom of automatic transmission / transaxle system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component repair work / service / replacement, programming and calibration carried out, fault code erased,

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Electrical repair tool kit. 1.4 Automatic transmission / transaxle electrical system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 1.5 Automatic transmission / transaxle electrical system fault troubleshooting procedure. • Intermittent fault. • Permanent fault.			test drive executed according to manufacturer diagnostic procedure. 1.5 automatic transmission / transaxle system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 1.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 1.2 Step-by-step approach followed, faults identified and resolved systematically. 1.3 Appropriate PPE complied according to safety requirement. 1.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Unknown fault. 1.6 Automatic transmission / transaxle electrical system fault rectification procedure: • Rescanning fault code. • Component repair. • Component service. • Component replacement. 1.7 Automatic transmission / transaxle electrical system testing procedure. • Static test. • Dynamic test. • Functionality test. 1.8 Type of automatic transmission /			according to manufacturer manual. 1.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 1.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 1.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	transaxle electrical system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 1.9 Automatic transmission / transaxle electrical system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC).			

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
2. Diagnose Continuous Variable Transmission (CVT) electrical system.	2.1 CVT electrical system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 2.2 CVT electrical system technical investigation information: • Vehicle CVT electrical system (layout, component, specification).	2.1 Review technical investigation and service information report. 2.2 Locate CVT mechatronic system fault. 2.3 Troubleshoot CVT system fault. 2.4 Rectify CVT system fault. 2.5 Test CVT system. 2.6 Produce CVT system diagnostic report.	<u>ATTITUDE</u> 2.1 Attention to detail while diagnosing CVT electrical system fault. 2.2 Systematic troubleshooting CVT electrical system issues. <u>SAFETY</u> 2.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 2.2 Proper handling of tools and equipment in accordance with safety regulations and	<u>COGNITIVE DOMAIN</u> 2.1 CVT electrical system safety explained. 2.2 CVT electrical system technical investigation information described. 2.3 CVT electrical system diagnostic tools application explained. 2.4 CVT electrical system fault identification explained. 2.5 CVT electrical system fault troubleshooting procedure listed. 2.6 CVT electrical system fault rectification procedure described. 2.7 CVT electrical system testing procedure identified.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 2.3 CVT electrical system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Hand tools. • Test lead kit. • Oil pressure gauge. • Soldering tool kit.		manufacturer requirement. 2.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 2.1 Proper disposal of hazardous materials in accordance with environmental regulations. 2.2 Minimize emissions and pollution in accordance with environmental regulations.	2.8 Type of CVT electrical system diagnostic report explained. 2.9 CVT electrical system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue) inspected and electrical components of CVT system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, CVT system wiring diagram interpreted, malfunction symptom of CVT system detected and data stream analysis interpreted according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Electrical repair tool kit. 2.4 CVT electrical system fault identification: • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 2.5 CVT electrical system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 2.6 CVT electrical system fault rectification procedure:			2.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 2.5 CVT system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 2.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 2.2 Step-by-step approach followed, faults identified and resolved systematically. 2.3 Appropriate PPE complied according to safety requirement. 2.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Rescanning fault code. • Component repair. • Component service. • Component replacement. 2.7 CVT electrical system testing procedure. • Static test. • Dynamic test. • Functionality test. 2.8 Type of CVT electrical system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report.			working on electrical systems according to manufacturer manual. 2.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 2.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 2.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Authority (JPJ) report procedure. 2.9 CVT electrical system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
3. Diagnose Dual Clutch Transmission (DCT) electrical system.	3.1 DCT electrical system safety: • Usage and application of PPE. • Work area preparation.	3.1 Review technical investigation and service information report. 3.2 Locate DCT mechatronic system fault.	<u>ATTITUDE</u> 3.1 Attention to detail while diagnosing DCT electrical system fault. 3.2 Systematic troubleshooting DCT	<u>COGNITIVE DOMAIN</u> 3.1 DCT electrical system safety explained. 3.2 DCT electrical system technical investigation information described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Tools and equipment handling. 3.2 DCT electrical system technical investigation information: • Vehicle DCT electrical system (type, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status.	3.3 Troubleshoot DCT electrical system fault. 3.4 Rectify DCT electrical system fault. 3.5 Test DCT electrical system. 3.6 Produce DCT electrical system diagnostic report.	electrical system issues. <u>SAFETY</u> 3.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 3.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 3.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 3.1 Proper disposal of hazardous materials in accordance with environmental regulations. 3.2 Minimize emissions and pollution in accordance with	3.3 DCT electrical system diagnostic tools application explained. 3.4 DCT electrical system fault identification explained. 3.5 DCT electrical system fault troubleshooting procedure listed. 3.6 DCT electrical system fault rectification procedure described. 3.7 DCT electrical system testing procedure identified. 3.8 Type of DCT electrical system diagnostic report explained. 3.9 DCT electrical system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms (shifting issue, wiring issue, control module issue, indicator issue, noise issue, vibration issue, shift by wire issue) inspected and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	3.3 DCT electrical system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Hand tools. • Test lead kit. • Oil pressure gauge. • DCT special service tool. • Soldering tool kit. • Electrical repair tool kit. 3.4 DCT electrical system fault identification: • Wiring diagram. • Waveform (sensors and actuators).		environmental regulations.	electrical components of DCT mechatronic system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, DCT electrical system wiring diagram interpreted, malfunction symptom of DCT electrical system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 3.4 Component repair work / service / replacement carried out, fault code erased, test drive executed according to manufacturer diagnostic procedure. 3.5 DCT electrical system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Manufacturers data specification. 3.5 DCT electrical system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 3.6 DCT electrical system fault rectification procedure: • Rescanning fault code. • Component repair. • Component service. • Component replacement. 3.7 DCT electrical system testing procedure.			<u>AFFECTIVE DOMAIN</u> 3.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 3.2 Step-by-step approach followed, faults identified and resolved systematically. 3.3 Appropriate PPE complied according to safety requirement. 3.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 3.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 3.6 Hazardous materials such as used oil, coolant, batteries, and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Static test. • Dynamic test. • Functionality test. 3.8 Type of DCT electrical system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 3.9 DCT electrical system diagnostic report format and procedure: • Type of report. • Type of fault.			other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 3.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			

Employability Skills

Core Abilities

- Please refer NCS- Core Abilities latest edition.

Social Values & Social Skills

- Please refer Handbook on Social Skills and Social Values in Technical Education and Vocational Training.

References for Learning Material Development

1	Bonnick, A. (2020). Automotive Powertrain Science and Technology (1st ed.). London: Routledge. ISBN 9780367331139.
2	Crouse, W. H. (1993). Automotive Mechanics (10th ed.). Career Education: New York. ISBN 9780028009438.
3	Denton, T. (2017). Automobile Mechanical and Electrical Systems (2nd ed.). Routledge. ISBN 978-0415725781.
4	Halderman, J. (2016). Automotive Electrical and Electronics (5th ed.). Upper Saddle River, NJ: Pearson. ISBN 9780134073644.
5	Stoakes, G. (2017). Automotive Oscilloscopes: Waveform Analysis. Graham Stoakes: United Kingdom. ISBN 9780992949266.

15.5 Perform automotive Advance Driver Assistance System (ADAS) diagnosis.

SECTION	(G) Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles		
GROUP	(452) Maintenance and Repair of Motor Vehicles		
AREA	Maintenance and Repair of Motor Vehicles		
NOSS TITLE	Automotive Electrical Diagnostic		
COMPETENCY UNIT TITLE	Perform automotive Advance Driver Assistance System (ADAS) diagnosis.		
LEARNING OUTCOMES	The learning outcomes of this competency are to enable the trainees to investigate, assess, troubleshoot, and calibrate complex ADAS components like radar, cameras, sensors, and control modules. It involves understanding how each ADAS feature (such as adaptive cruise control, lane-keeping assist, and automatic emergency braking) functions and integrates with other vehicle systems, as well as knowing how to use specialized diagnostic tools to retrieve and interpret error codes and sensor data. Upon completion of this competency unit, trainees should be able to: 1. Diagnose radar system. 2. Diagnose camera system. 3. Diagnose ultrasonic sensor system. 4. Diagnose Light Detection and Ranging (LiDAR) system.		
TRAINING PREREQUISITE (SPECIFIC)	Not Available.		
CU CODE	G452-011-4:2025-C05	NOSS LEVEL	Four (4)

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
1. Diagnose radar system.	1.1 Radar system safety:	1.1 Review technical investigation and service information report.	ATTITUDE 1.1 Attention to detail while diagnosing radar system fault.	COGNITIVE DOMAIN 1.1 Radar system safety explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 1.2 Radar system technical investigation information: • Vehicle radar system (layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation.	1.2 Locate radar system fault. 1.3 Troubleshoot radar system fault. 1.4 Rectify radar system fault. 1.5 Test radar system. 1.6 Produce radar system diagnostic report.	1.2 Systematic troubleshooting radar system issues. <u>SAFETY</u> 1.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 1.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 1.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 1.1 Proper disposal of hazardous materials in accordance with environmental regulations. 1.2 Minimize emissions and pollution in	1.2 Radar system technical investigation information described. 1.3 Radar system diagnostic tools application explained. 1.4 Radar system fault identification explained. 1.5 Radar system fault troubleshooting procedure listed. 1.6 Radar system fault rectification procedure described. 1.7 Radar system testing procedure identified. 1.8 Type of radar system diagnostic report explained. 1.9 Radar system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 1.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 1.2 Symptoms inspected (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue,

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Software version status. 1.3 Radar system diagnostic tools application: • Scan tools. • Digital multimeter. • Oscilloscope. • Hand tools. • Test lead kit. • Radar calibration tool kit. 1.4 Radar system fault identification: • Log file. • Wiring diagram. • Waveform (sensors and actuators). • Manufacturers data specification. 1.5 Radar system fault		accordance with environmental regulations.	indicator, system error, cruise control issue, software issue) and electrical components of radar system tested according to manufacturer diagnostic procedure. 1.3 Error code identified, radar system wiring diagram interpreted, malfunction symptom of radar system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 1.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 1.5 Radar system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 1.6 Summary report / technical service bulletin / part bulletin / warranty report prepared.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 1.6 Radar system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 1.7 Radar system testing procedure. • Static test. • Dynamic test. • Functionality test. 1.8 Type of radar system diagnostic report:			AFFECTIVE DOMAIN 1.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 1.2 Step-by-step approach followed, faults identified and resolved systematically. 1.3 Appropriate PPE complied according to safety requirement. 1.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 1.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 1.6 Hazardous materials such as used oil, coolant, batteries, and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 1.9 Radar system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification.			other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 1.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Recommendation. • Approval by superior.			
2. Diagnose camera system.	2.1 Camera system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 2.2 Camera system technical investigation information: • Vehicle camera system (types, layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history.	2.1 Review technical investigation and service information report. 2.2 Locate camera system fault. 2.3 Troubleshoot camera system fault. 2.4 Rectify camera system fault. 2.5 Test camera system. 2.6 Produce camera system diagnostic report.	<u>ATTITUDE</u> 2.1 Attention to detail while diagnosing camera system fault. 2.2 Systematic troubleshooting camera system issues. <u>SAFETY</u> 2.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 2.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 2.3 Maintain a safe and organized work area in accordance with safety regulations.	<u>COGNITIVE DOMAIN</u> 2.1 Camera system safety explained. 2.2 Camera system technical investigation information described. 2.3 Camera system diagnostic tools application explained. 2.4 Camera system fault identification explained. 2.5 Camera system fault troubleshooting procedure listed. 2.6 Camera system fault rectification procedure described. 2.7 Camera system testing procedure identified. 2.8 Type of camera system diagnostic report explained. 2.9 Camera system diagnostic report format and procedure described.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Functionality test result. • Live data stream. • Software version status. 2.3 Camera system diagnostic tools application: • Scan tools. • Digital multimeter. • Hand tools. • Test lead kit. • Camera calibration tool kit. 2.4 Camera system fault identification: • Log file. • Wiring diagram. • Sensors test. • Manufacturers data specification.		<u>ENVIRONMENT</u> 2.1 Proper disposal of hazardous materials in accordance with environmental regulations. 2.2 Minimize emissions and pollution in accordance with environmental regulations.	<u>PSYCHOMOTOR DOMAIN</u> 2.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 2.2 Symptoms inspected (monitoring issue, sensing issue, wiring issue, indicator, sensor issue, false warning issue) and electrical components of camera system tested according to manufacturer diagnostic procedure. 2.3 Error code identified, camera system wiring diagram interpreted, malfunction symptom of camera detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 2.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.5 Camera system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 2.6 Camera system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 2.7 Camera system testing procedure. • Static test. • Dynamic test. • Functionality test.			2.5 Camera re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 2.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 2.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 2.2 Step-by-step approach followed, faults identified and resolved systematically. 2.3 Appropriate PPE complied according to safety requirement. 2.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 2.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	2.8 Type of camera system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 2.9 Camera system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result.			lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 2.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 2.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Result analysis. • Justification. • Recommendation. • Approval by superior.			
3. Diagnose ultrasonic sensor system.	3.1 Ultrasonic sensor system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 3.2 Ultrasonic system technical investigation information: • Vehicle ultrasonic sensor system (layout, component, specification).	3.1 Review technical investigation and service information report. 3.2 Locate ultrasonic sensor system fault. 3.3 Troubleshoot ultrasonic sensor system fault. 3.4 Rectify ultrasonic sensor system fault. 3.5 Test ultrasonic sensor system. 3.6 Produce ultrasonic sensor system diagnostic report.	<u>ATTITUDE</u> 3.1 Attention to detail while diagnosing ultrasonic sensor system fault. 3.2 Systematic troubleshooting ultrasonic sensor system issues. <u>SAFETY</u> 3.1 Use of Personal Protective Equipment (PPE) in accordance with safety regulations. 3.2 Proper handling of tools and equipment in accordance with safety regulations and	<u>COGNITIVE DOMAIN</u> 3.1 Ultrasonic sensor system safety explained. 3.2 Ultrasonic sensor system technical investigation information described. 3.3 Ultrasonic sensor system diagnostic tools application explained. 3.4 Ultrasonic sensor system fault identification explained. 3.5 Ultrasonic sensor system fault troubleshooting procedure listed. 3.6 Ultrasonic sensor system fault rectification procedure described. 3.7 Ultrasonic sensor system testing procedure identified.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 3.3 Ultrasonic sensor system diagnostic tools application: • Scan tools. • Data logger (D-logger). • Digital multimeter. • Oscilloscope. • Hand tools. • Test lead kit. 3.4 Ultrasonic sensor system fault identification:		manufacturer requirement. 3.3 Maintain a safe and organized work area in accordance with safety regulations. ENVIRONMENT 3.1 Proper disposal of hazardous materials in accordance with environmental regulations. 3.2 Minimize emissions and pollution in accordance with environmental regulations.	3.8 Type of ultrasonic sensor system diagnostic report explained. 3.9 Ultrasonic sensor system diagnostic report format and procedure described. PSYCHOMOTOR DOMAIN 3.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 3.2 Symptoms inspected (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue, indicator, system error, cruise control issue, software issue) and electrical components of ultrasonic sensor system tested according to manufacturer diagnostic procedure. 3.3 Error code identified, ultrasonic sensor system wiring diagram interpreted, malfunction symptom of ultrasonic sensor system detected and data stream analysis interpreted according

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Log file. • Wiring diagram. • Waveform (sensors). • Manufacturers data specification. 3.5 Ultrasonic sensor system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault. 3.6 Ultrasonic sensor system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement.			to manufacturer diagnostic procedure. 3.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 3.5 Ultrasonic sensor system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 3.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 3.1 Every component carefully examined, error codes read accurately, and connections and readings double-checked. 3.2 Step-by-step approach followed, faults identified and resolved systematically. 3.3 Appropriate PPE complied according to safety requirement.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	3.7 Ultrasonic sensor system testing procedure. • Static test. • Dynamic test. • Functionality test. 3.8 Type of ultrasonic sensor system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign. • Warranty report. • Authority (JPJ) report procedure. 3.9 Ultrasonic sensor system diagnostic report			3.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 3.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 3.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and handled by certified/licensed recycling or disposal facility/transporter. 3.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code-(DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			
4. Diagnose Light Detection and Ranging (LiDAR) system.	4.1 LiDAR system safety: • Usage and application of PPE. • Work area preparation. • Tools and equipment handling. 4.2 LiDAR system technical	4.1 Review technical investigation and service information report. 4.2 Locate LiDAR system fault. 4.3 Troubleshoot LiDAR system fault. 4.4 Rectify LiDAR system fault. 4.5 Test LiDAR system.	<u>ATTITUDE</u> 4.1 Attention to detail while diagnosing LiDAR system fault. 4.2 Systematic troubleshooting LiDAR system issues. <u>SAFETY</u> 4.1 Use of Personal Protective Equipment (PPE) in accordance	<u>COGNITIVE DOMAIN</u> 4.1 LiDAR system safety explained. 4.2 LiDAR system technical investigation information described. 4.3 LiDAR system diagnostic tools application explained. 4.4 LiDAR system fault identification explained.

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	investigation information: • Vehicle LiDAR system (layout, component, specification). • Type of complaints. • Fault symptoms. • Fault history. • Functionality test result. • Live data stream. • Waveform interpretation. • Software version status. 4.3 LiDAR system diagnostic tools application: • Scan tools. • Data logger (D-logger).	4.6 Produce LiDAR system diagnostic report.	with safety regulations. 4.2 Proper handling of tools and equipment in accordance with safety regulations and manufacturer requirement. 4.3 Maintain a safe and organized work area in accordance with safety regulations. <u>ENVIRONMENT</u> 4.1 Proper disposal of hazardous materials in accordance with environmental regulations. 4.2 Minimize emissions and pollution in accordance with environmental regulations.	4.5 LiDAR system fault troubleshooting procedure listed. 4.6 LiDAR system fault rectification procedure described. 4.7 LiDAR system testing procedure identified. 4.8 Type of LiDAR system diagnostic report explained. 4.9 LiDAR system diagnostic report format and procedure described. <u>PSYCHOMOTOR DOMAIN</u> 4.1 Customer complaint, technician findings, symptom, technical report, evidence, service history analysed. 4.2 Symptoms (wiring issue, sensor module issue, distance reading, weather issue, inconsistent issue, indicator, system error, cruise control issue, software issue, physical and mechanical issue) inspected and electrical components of LiDAR system tested according to

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Digital multimeter. • Oscilloscope. • Test lead kit. • Hand tools. • LiDAR sensor diagnostic kit. 4.4 LiDAR system fault identification: • Log file. • Wiring diagram. • Waveform (sensors). • Manufacturers data specification. 4.5 LiDAR system fault troubleshooting procedure. • Intermittent fault. • Permanent fault. • Unknown fault.			manufacturer diagnostic procedure. 4.3 Error code identified, LiDAR system wiring diagram interpreted, malfunction symptom of LiDAR system detected and data stream analysis interpreted according to manufacturer diagnostic procedure. 4.4 Component service / replacement, programming and calibration carried out, fault code erased, test drive (static/dynamic) executed according to manufacturer diagnostic procedure. 4.5 LiDAR system re-diagnosed and its functionality confirmed according to manufacturer diagnostic procedure. 4.6 Summary report / technical service bulletin / part bulletin / warranty report prepared. <u>AFFECTIVE DOMAIN</u> 4.1 Every component carefully examined, error codes read

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	4.6 LiDAR system fault rectification procedure: • Rescanning fault code. • Component service. • Component replacement. 4.7 LiDAR system testing procedure. • Static test. • Dynamic test. • Functionality test. 4.8 Type of LiDAR system diagnostic report: • Summary report. • Technical service bulletin. • Part bulletin. • Recall campaign.			accurately, and connections and readings double-checked. 4.2 Step-by-step approach followed, faults identified and resolved systematically. 4.3 Appropriate PPE complied according to safety requirement. 4.4 Good working condition tools and diagnostic equipment used correctly, and the vehicle battery disconnected when working on electrical systems according to manufacturer manual. 4.5 Workspace clean and free from clutter to reduce tripping hazards. Jack stands or vehicle lifts properly used to ensure the vehicle is stable when working underneath it and hazardous materials secured and clearly labelled to prevent accidental spills or exposure. 4.6 Hazardous materials such as used oil, coolant, batteries, and other automotive fluids collected and disposed of in designated containers and

WORK ACTIVITIES	RELATED KNOWLEDGE	RELATED SKILLS	ATTITUDE/ SAFETY/ ENVIRONMENT	ASSESSMENT CRITERIA
	• Warranty report. • Authority (JPJ) report procedure. 4.9 LiDAR system diagnostic report format and procedure: • Type of report. • Type of fault. • Diagnostic Trouble Code- (DTC). • Test result. • Result analysis. • Justification. • Recommendation. • Approval by superior.			handled by certified/licensed recycling or disposal facility/transporter. 4.7 Engines turned off when not in use and avoiding prolonged idling and proper equipment used to capture exhaust fumes in a workshop setting.

Employability Skills

Core Abilities

- Please refer NCS- Core Abilities latest edition.

Social Values & Social Skills

- Please refer Handbook on Social Skills and Social Values in Technical Education and Vocational Training.

References for Learning Material Development

1	Crouse, W. H. (1993). Automotive Mechanics (10th ed.). Career Education: New York. ISBN 9780028009438.
2	Denton, T. (2019). Automated Driving and Driver Assistance Systems. Routledge. ISBN 9780367265595.
3	Stoakes, G. (2017). Automotive Oscilloscopes: Waveform Analysis. Graham Stoakes: United Kingdom. ISBN 9780992949266.

16. Delivery Mode

The following are the **recommended** training delivery modes: -

KNOWLEDGE	SKILL
• Lecture • Group discussion • E-learning, self-paced • E-learning, facilitate • Case study or Problem based learning (PBL) • Self-paced learning, non-electronic • One-on-one tutorial • Shop talk • Seminar	• Demonstration • Simulation • Project • Scenario based training (SBT) • Role play • Coaching • Observation • Mentoring

Skills training and skills assessment of trainees should be implemented in accordance with TEM requirements and actual situation.

17. Tools, Equipment and Materials (TEM)

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

CU	CU CODE	COMPETENCY UNIT TITLE
C01	G452-011-4:2025-C01	Perform automotive Engine Management System (EMS) diagnosis.
C02	G452-011-4:2025-C02	Perform automotive body electrical system diagnosis.
C03	G452-011-4:2025-C03	Perform automotive chassis electrical diagnosis.
C04	G452-011-4:2025-C04	Perform automotive powertrain system diagnosis.
C05	G452-011-4:2025-C05	Perform automotive Advance Driver Assistance System (ADAS) diagnosis.

* Items listed refer to TEM's **minimum requirement** for skills delivery only.

NO.	ITEM*	RATIO (TEM : Trainees or AR = As Required)				
		C01	C02	C03	C04	C05
A. Tools						
1	AC compressor clutch remover.		1:25			
2	AC disconnection tool set.		1:25			
3	AC/DC clamp meter.	1:5	1:5	1:5	1:5	1:5
4	Breakout box.	1:25	1:25	1:25		
5	Camera calibration tool kit.					1:25
6	Compression test kit.	1:5				
7	Data logger (D-logger).	1:25	1:25	1:25	1:25	1:25
8	DCT special service tool.				1:25	
9	Digital multimeter.	1:1	1:1	1:1	1:1	1:1
10	Digital thermometer.		1:5			
11	Electrical expansion valve test kit.		1:25			
12	Electrical repair tool kit.	1:25	1:25		1:25	
13	Fuel pressure gauge.	1:5				

14	Torque wrench set.	1:25	1:25	1:25	1:25	1:25
15	Hand toolbox set.	1:1	1:1	1:1	1:1	1:1
16	Tools set trolley.	1:5	1:5	1:5	1:5	1:5
17	Key transponder detector kit.		1:5			
18	Leak test kit (UV light, adapter, UV glass, electronic leak detector).		1:25			
19	LiDAR sensor diagnostic kit.					1:25
20	Manifold gauge.		1:5			
21	Noid light tester.	1:5				
22	Oil pressure gauge.				1:25	
23	Piston brake retractor kit.			1:25		
24	Soldering tool kit.	1:5	1:5		1:5	
25	Test lead kit.	1:25	1:25	1:25	1:25	1:25
26	Tyre pressure gauge.			1:25		
27	Vacuum pump.	1:25				
28	Belt tension gauge.		1:25			
B. Equipment						
1	Air conditioning recovery machine.		1:25			
2	Battery charger / Power backup.	1:25	1:25	1:25	1:25	1:25
3	Battery tester.	1:25	1:25	1:25	1:25	1:25
4	Brake bleeding machine.			1:25		
5	Emission analyzer.	1:25				
6	Headlight alignment machine.		1:25			
7	Ignition coil analyzer.	1:25				
8	Injector tester.	1:25				
9	Oscilloscope with special service tool.	1:25	1:25	1:25	1:25	1:25
10	Radar calibration tool kit.					1:25
11	Refrigerant analyzer.		1:25			
12	Scan tools.	1:5	1:5	1:5	1:5	1:5
13	Sensor / actuator signal simulator.	1:25				
14	SRS test kit.		1:25			

15	TPMS diagnostic tools.			1:25		
16	Tyre changer.			1:25		
17	Wheel alignment machine.			1:25		
18	Wind speed meter.		1:25			
19	Floor jack.	1:5	1:5	1:5	1:5	1:5
20	Safety stand.	1:5	1:5	1:5	1:5	1:5
21	Automatic vehicle	1:25	1:25	1:25	1:25	1:25
22	Manual vehicle	1:25	1:25	1:25	1:25	1:25
23	Vehicle Protective Equipment (VPE)	1:25	1:25	1:25	1:25	1:25
24	Personnel Protective Equipment (PPE)	1:1	1:1	1:1	1:1	1:1
25	Hoist	1:25	1:25	1:25	1:25	1:25
26	Air compressor	1:25	1:25	1:25	1:25	1:25
C. Materials						
1	Job order sample.	AR	AR	AR	AR	AR
2	Customer handling sample.	AR	AR	AR	AR	AR
3	Chronology of event sample.	AR	AR	AR	AR	AR
4	Warranty term & condition sample.	AR	AR	AR	AR	AR
5	Service manual.	1:25	1:25	1:25	1:25	1:25
6	Workshop manual.	1:25	1:25	1:25	1:25	1:25
7	Sample report.	AR	AR	AR	AR	AR
8	Sample checklist.	AR	AR	AR	AR	AR
9	Sample of workshop housekeeping manual	1:25	1:25	1:25	1:25	1:25
10	Soap solution	AR	AR	AR	AR	AR
11	R134a refrigerant		AR			
12	Compressor oil		AR			
13	UV liquid		AR			
14	Brake fluid			AR		
15	Transmission fluid (ATF/ DCT / CVT)				AR	
16	Contact cleaner	AR	AR	AR	AR	AR
17	Sensor and radar liquid cleaner					AR
18	Cotton rag	AR	AR	AR	AR	AR

19	Insulation tape	AR	AR	AR	AR	AR
20	Wire tape	AR	AR	AR	AR	AR
21	Degreaser		AR	AR	AR	
22	O-ring repair kit		AR			

18. Competency Weightage

The following table shows the percentage of training priorities based on consensus made by the Standard Development Committee (SDC).

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

CU CODE	COMPETENCY UNIT TITLE	COMPETENCY UNIT WEIGHTAGE	WORK ACTIVITIES	WORK ACTIVITIES WEIGHTAGE
G452-011-4:2025-C01	Perform automotive Engine Management System (EMS) diagnosis.	25%	1. Diagnose ignition system.	20%
			2. Diagnose starting system.	15%
			3. Diagnose charging system.	15%
			4. Diagnose electric fuel delivery system.	25%
			5. Diagnose engine control system	25%
G452-011-4:2025-C02	Perform automotive body electrical system diagnosis.	30%	1. Diagnose lighting and instrument panel system.	20%
			2. Diagnose washer and wiper system.	10%
			3. Diagnose power window and mirror system.	10%
			4. Diagnose infotainment system.	20%
			5. Diagnose security and lock system.	15%
			6. Diagnose Supplemental Restraint System (SRS).	10%
			7. Diagnose Electrical Heating, Ventilation and Air Conditioning system (HVAC).	15%
G452-011-4:2025-C03	Perform automotive chassis electrical diagnosis.	15%	1. Diagnose Electric Power Steering (EPS) system.	25%

			2. Diagnose Anti-lock Braking System (ABS).	25%
			3. Diagnose Electronic Parking Brake (EPB).	15%
			4. Diagnose Electronic Stability Programme (ESP) system.	25%
			5. Diagnose Tyre Pressure Monitoring System (TPMS).	10%
G452-011-4:2025-C04	Perform automotive powertrain system diagnosis.	15%	1. Diagnose automatic transmission / transaxle electrical system.	40%
			2. Diagnose Continuous Variable Transmission (CVT) electrical system.	25%
			3. Diagnose Dual Clutch Transmission (DCT) electrical system.	35%
G452-011-4:2025-C05	Perform automotive Advance Driver Assistance System (ADAS) diagnosis.	15%	1. Diagnose radar system.	30%
			2. Diagnose camera system.	30%
			3. Diagnose ultrasonic sensor system.	15%
			4. Diagnose Light Detection and Ranging (LiDAR) system.	25%
TOTAL PERCENTAGE (CORE COMPETENCY)		= 100%		

APPENDICES

NATIONAL OCCUPATIONAL SKILLS STANDARD (NOSS) FOR:

AUTOMOTIVE ELECTRICAL DIAGNOSTIC

LEVEL 4

19. Appendices

19.1 Appendix A: Competency Profile Chart For Teaching & Learning (CPC_{PdP})

i. CU to CU_{PdP} Correlation

SECTION	(G) WHOLESALE AND RETAIL TRADE; REPAIR OF MOTOR VEHICLES AND MOTORCYCLES		
GROUP	(452) MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
AREA	MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
NOSS TITLE	AUTOMOTIVE ELECTRICAL DIAGNOSTIC		
NOSS LEVEL	FOUR (4)	NOSS CODE	G452-011-4:2025

CU CODE	CU TITLE	CU_{PdP} TITLE For Teaching & Learning
G452-011-4:2025-C01	PERFORM AUTOMOTIVE ENGINE MANAGEMENT SYSTEM (EMS) DIAGNOSIS	AUTOMOTIVE ENGINE MANAGEMENT SYSTEM (EMS) DIAGNOSTIC
G452-011-4:2025-C02	PERFORM AUTOMOTIVE BODY ELECTRICAL SYSTEM DIAGNOSIS	AUTOMOTIVE BODY ELECTRICAL SYSTEM DIAGNOSTIC
G452-011-4:2025-C03	PERFORM AUTOMOTIVE CHASSIS ELECTRICAL DIAGNOSIS	AUTOMOTIVE CHASSIS ELECTRICAL DIAGNOSTIC
G452-011-4:2025-C04	PERFORM AUTOMOTIVE POWERTRAIN SYSTEM DIAGNOSIS	AUTOMOTIVE POWERTRAIN SYSTEM DIAGNOSTIC
G452-011-4:2025-C05	PERFORM AUTOMOTIVE ADVANCE DRIVER ASSISTANCE SYSTEM (ADAS) DIAGNOSIS	AUTOMOTIVE ADVANCE DRIVER ASSISTANCE SYSTEM (ADAS) DIAGNOSTIC

ii. Competency Profile Chart for Teaching & Learning (CPC$_{PdP}$)

SECTION	(G) WHOLESALE AND RETAIL TRADE; REPAIR OF MOTOR VEHICLES AND MOTORCYCLES		
GROUP	(452) MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
AREA	MAINTENANCE AND REPAIR OF MOTOR VEHICLES		
NOSS TITLE	AUTOMOTIVE ELECTRICAL DIAGNOSTIC		
NOSS LEVEL	FOUR (4)	NOSS CODE	G452-011-4:2025

	↤COMPETENCY UNIT↦	↤WORK ACTIVITIES↦			
CORE	AUTOMOTIVE ENGINE MANAGEMENT SYSTEM (EMS) DIAGNOSTIC G452-011-4:2025-C01	DIAGNOSE IGNITION SYSTEM G452-011-4:2025-C01-W01	DIAGNOSE STARTING SYSTEM G452-011-4:2025-C01-W02	DIAGNOSE CHARGING SYSTEM G452-011-4:2025-C01-W03	DIAGNOSE ELECTRIC FUEL DELIVERY SYSTEM G452-011-4:2025-C01-W04
		DIAGNOSE ENGINE CONTROL SYSTEM G452-011-4:2025-C01-W05			

	↤COMPETENCY UNIT↦	↤WORK ACTIVITIES↦			
CORE	AUTOMOTIVE BODY ELECTRICAL SYSTEM DIAGNOSTIC G452-011-4:2025-C02	DIAGNOSE LIGHTING AND INSTRUMENT PANEL SYSTEM G452-011-4:2025-C02-W01	DIAGNOSE WASHER AND WIPER SYSTEM G452-011-4:2025-C02-W02	DIAGNOSE POWER WINDOW AND MIRROR SYSTEM G452-011-4:2025-C02-W03	DIAGNOSE INFOTAINMENT SYSTEM G452-011-4:2025-C02-W04
		DIAGNOSE SECURITY AND LOCK SYSTEM G452-011-4:2025-C02-W05	DIAGNOSE SUPPLEMENTAL RESTRAINT SYSTEM (SRS) G452-011-4:2025-C02-W06	DIAGNOSE ELECTRICAL HEATING, VENTILATION AND AIR CONDITIONING SYSTEM (HVAC) G452-011-4:2025-C02-W07	
	AUTOMOTIVE CHASSIS ELECTRICAL DIAGNOSTIC G452-011-4:2025-C03	DIAGNOSE ELECTRIC POWER STEERING (EPS) SYSTEM G452-011-4:2025-C03-W01	DIAGNOSE ANTI-LOCK BRAKING SYSTEM (ABS). G452-011-4:2025-C03-W02	DIAGNOSE ELECTRONIC PARKING BRAKE (EPB). G452-011-4:2025-C03-W03	DIAGNOSE ELECTRONIC STABILITY PROGRAMME (ESP) SYSTEM G452-011-4:2025-C03-W04

	↤COMPETENCY UNIT↦	↤WORK ACTIVITIES↦			
CORE		DIAGNOSE TYRE PRESSURE MONITORING SYSTEM (TPMS) G452-011-4:2025-C03-W05			
CORE	AUTOMOTIVE POWERTRAIN SYSTEM DIAGNOSTIC G452-011-4:2025-C04	DIAGNOSE AUTOMATIC TRANSMISSION / TRANSAXLE ELECTRICAL SYSTEM G452-011-4:2025-C04-W01	DIAGNOSE CONTINUOUS VARIABLE TRANSMISSION (CVT) ELECTRICAL SYSTEM G452-011-4:2025-C04-W02	DIAGNOSE DUAL CLUTCH TRANSMISSION (DCT) ELECTRICAL SYSTEM G452-011-4:2025-C04-W03	
CORE	AUTOMOTIVE ADVANCE DRIVER ASSISTANCE SYSTEM (ADAS) DIAGNOSTIC G452-011-4:2025-C05	DIAGNOSE RADAR SYSTEM G452-011-4:2025-C05-W01	DIAGNOSE CAMERA SYSTEM G452-011-4:2025-C05-W02	DIAGNOSE ULTRASONIC SENSOR SYSTEM G452-011-4:2025-C05-W03	DIAGNOSE LIGHT DETECTION AND RANGING (LIDAR) SYSTEM G452-011-4:2025-C05-W04

Notes:

CPC_{PdP} is meant to be used in Teaching and Learning context which is generated by conversion of the action verb in the CU Title to a noun in the CU_{PdP} Title from the given CPC sets.

19.2 Appendix B: Element Content Weightage

OSH - OCCUPATIONAL SAFETY AND HEALTH
SD - SUSTAINABLE DEVELOPMENT
M&A - MANAGEMENT AND ADMINISTRATION
IT - INDUSTRY TECHNOLOGICAL ADVANCES

AUTOMOTIVE ELECTRICAL DIAGNOSTIC
LEVEL 4

CU CODE	CU TITLE	ELEMENT CONTENT WEIGHTAGE			
		OSH	SD	M&A	IT
G452-011-4:2025-C01	Perform automotive Engine Management System (EMS) diagnosis	15%	25%	25%	25%
G452-011-4:2025-C02	Perform automotive body electrical system diagnosis	20%	20%	20%	15%
G452-011-4:2025-C03	Perform automotive chassis electrical diagnosis	15%	15%	15%	15%
G452-011-4:2025-C04	Perform automotive powertrain system diagnosis	25%	15%	15%	20%
G452-011-4:2025-C05	Perform automotive Advance Driver Assistance System (ADAS) diagnosis	25%	25%	25%	25%

CU CODE	CU TITLE	ELEMENT CONTENT WEIGHTAGE			
		OSH	SD	M&A	IT
TOTAL ELEMENT CONTENT WEIGHTAGE		100%	100%	100%	100%
NOTES		C05 often involve working with advanced sensors (e.g., radars, LiDAR, and cameras) and high-voltage components, which can pose electrical hazards. Calibration and testing frequently require safe handling procedures to prevent accidents, especially during live vehicle testing. C03 involve mechanical components operating under high pressure or temperature. Diagnosing these	C01 contributes to sustainable development by optimizing engine performance, improving fuel efficiency, and reducing harmful emissions. This aligns with environmental sustainability goals by minimizing the vehicle's carbon footprint. C05 - ADAS systems enhance safety and support sustainable development by reducing accidents and promoting efficient driving behaviors.	C05 - ADAS systems are complex and require integration across multiple vehicle subsystems, extensive documentation, coordination with software updates, compliance with safety regulations, and collaboration with OEMs (Original Equipment Manufacturers). This involves significant administrative oversight to ensure proper diagnostics and system recalibrations.	C01 integrates advanced technologies such as electronic fuel injection, variable valve timing, turbocharging, and hybrid or electric powertrain controls. The system often incorporates sophisticated sensors and actuators that are critical for optimizing engine performance, emissions reduction, and compliance with evolving regulations.

CU CODE	CU TITLE	ELEMENT CONTENT WEIGHTAGE			
		OSH	SD	M&A	IT
		systems requires strict adherence to OSH standards to avoid risks such as injuries from moving parts.	Additionally, these systems are foundational for autonomous and electric vehicles, which are pivotal for sustainable transportation solutions.	C01- interacts with various components, requiring detailed coordination, accurate record-keeping, and adherence to emission standards. Managing data from diagnostic tools and ensuring that repairs align with manufacturer guidelines also adds an administrative dimension.	C05 heavily relies on cutting-edge technologies like radar, LiDAR, cameras, ultrasonic sensors, and V2X (vehicle-to-everything) communication.